Supplementary Material: A Frequentist Framework for Estimating the Education Practices and Skill Probability Index (EPSPI)

Vivek Upadhyay[1,2] and **Amaresh Chakrabarti**[1]

[1]Department of Design and Manufacturing, Indian Institute of Science, Bengaluru, India

[2]Corresponding author: viveku@iisc.ac.in

1 Introduction

This supplementary article accompanies the main paper titled "A Frequentist Framework for Estimating the Education Practices and Skill Probability Index (EPSPI)". It is intended to enhance readability by providing additional methodological details and clarifications that support the primary analysis.

The appendix presents consolidated, ready-to-use formulas, parameter tables, and summary results, offering a concise reference for readers. It is designed to facilitate practical application by enabling practitioners to directly utilise the reported formulations and tabulated values. This article makes the following contributions:

(i) First, we provide a formal probabilistic definition of EPSPI, establishing its theoretical basis in conditional probability and its interpretability as a scalar alignment measure.

(ii) Second, we propose a three-layer architecture - line-level (or utterance-level), episode aggregate level, and episode evaluation level - that bridges the gap between fine-grained discourse annotation and coarser instructional evaluation.

(iii) Third, we develop and compare eleven frequentist approach-based estimation methods organised along three methodological axes: direct estimation, dimensionality reduction, and predictive modelling.

(iv) Fourth, we introduce a comprehensive evaluation framework comprising 7 metrics, cross-validated Pearson r, in-sample Train Pearson r, calibrated and raw explained variance (Exp.Var(cal) and Exp.Var(raw)), Cross-Validate Mean Squared Error, AIC, and BIC that together assess predictive validity, calibration, and model parsimony.

(v) Fifth, we conduct a leverage analysis that identifies the structural drivers of the performance ceiling, demonstrating that episode-level influence rather than model complexity is the binding constraint on out-of-sample prediction in this corpus.

(vi) Finally, we demonstrate that the observed episode skill rate $\bar{\mathrm{Y}}_e$, the proportion of skill-present observations across all utterances and skills within an episode constitutes a direct, model-independent empirical index of productive teaching activity, and we show how its decomposition into predicted and residual components supports both diagnostic interpretation and prospective deployment in teacher feedback contexts.

2 Data, Notation and EPSPI Probabilistic Formulation

2.1 Corpus and Variables

The corpus consists of $N = 19,181$ classroom utterance lines and teacher excerpts drawn from 49 episodes: Episodes 1–30 are Science lessons (grades 6–12), Episodes 31–48 are Mathematics lessons (grades 6–12), and Episode 49 is a vignette containing teacher excerpts from a grade-7 science lesson. Learning theories are referred to as Educational Practices (EP) in this study. Each line n is annotated with binary EP presence indicators and binary skill presence indicators:

- $X_{ni} \in \{0, 1\}$: EP i is active in line n, $i \in \{1, \ldots, P\}$ (Behaviourism, Cognitivism, Constructivism; $P = 3$).
- $Y_{nj} \in \{0, 1\}$: Skill j is present in line n, $j \in \{1, \ldots, K\}$ (Problem Solving, Creativity, Collaborative Learning, Critical Thinking; $K = 4$).
- $e_n \in \{1, \ldots, 49\}$: Episode of line n.
- $n_e = \{n : e_n = e\}$: The set of all lines n whose episode label equals e.

2.2 Marginal, Joint, and Conditional Probabilities

For a randomly selected line n, the marginal probability of EP_i being active and the marginal probability of skill S_j being present are:

$$P(EP_i) = \frac{\sum_{n=1}^{N} X_{ni}}{N}, \quad P(S_j) = \frac{\sum_{n=1}^{N} Y_{nj}}{N} \tag{1}$$

1	Global encoding	Speaker	Transcript	Time stamp		Learning Theories			21st Century Skills			
2					Behaviourism	Cognitivism	Constructivism	Problem Solving	Creativity	Collaborative Learning	Critical Thinking	Episode
3	Preassessment	Speaker1	you want to come	00:00:04	1	0	3	0	0	3	0	1
4			nidhi you come here 5 girls come herer	00:00:09	1	0	3	0	0	3	0	1
5			4 girls come here	00:00:12	1	0	3	0	0	3	0	1
6			you come here	00:00:18	1	0	3	0	0	3	0	1
7			come go there	00:00:21	1	0	3	0	0	3	0	1
8			you go there	00:00:24	1	0	3	0	0	3	0	1
9			you come how many girls	00:00:28	1	0	3	0	0	3	0	1
10			1234567 12345	00:00:31	1	0	3	0	0	3	0	1
11			12345	00:00:34	1	0	3	0	0	3	0	1
12			12345	00:00:37	1	0	3	0	0	3	0	1
13			now here we are 12345	00:00:41	1	0	3	0	0	3	0	1
14			now what you have to do	00:00:44	1	0	3	0	0	3	0	1
15			first thing in last class we have learned	00:00:48	1	0	3	0	0	3	0	1
16			herbs shurbs and tree	00:00:51	1	0	3	0	0	3	0	1
17			so here one group go this side	00:00:54	1	0	3	0	0	3	0	1
18			boy will go this side and girls will go this side	00:00:56	1	0	3	0	0	3	0	1
19			and each group will identify	00:00:59	1	0	3	0	0	3	0	1
20			the shrub and herb	00:01:04	1	0	3	0	0	3	0	1
21			what you have to identify these are the trees	00:01:08	1	0	3	0	0	3	0	1
22			but you have to find out	00:01:11	1	0	3	0	0	3	0	1
23			herb and shrub(in unison) later on later on	00:01:15	1	0	3	0	0	3	0	1
24			after getting this the shrub and this the	00:01:18	1	0	3	0	0	3	0	1

Figure 1: Sample of encoding process for variables Learning theories and 21st-century skills

The joint probability of EP_i and S_j co-occurring in the same line is:

$$P(EP_i \cap S_j) = \frac{\sum_{n=1}^{N} X_{ni} Y_{nj}}{N} \tag{2}$$

The conditional probability of skill S_j given that EP_i is active follows directly:

$$P(S_j | EP_i) = \frac{P(EP_i \cap S_j)}{P(EP_i)} = \frac{\sum_{n=1}^{N} X_{ni} Y_{nj}}{\sum_{n=1}^{N} X_{ni}} \tag{3}$$

Laplace smoothing: Since constructivism is absent from many individual episodes ($\sum_{\{n:e_n=e\}} X_{n,\mathrm{Con}} = 0$ for most e), the denominator of Equation (3) is zero. To resolve this, we use Laplace (additive) smoothing with parameters $\alpha = 1$ and $K = 4$, adding a small pseudo-count to both numerator and denominator. The smoothed estimate, denoted θ_{ij}, is:

$$P(S_j | EP_i) = \theta_{ij} = \frac{\sum_{n=1}^{N} X_{ni} Y_{nj} + \alpha}{\sum_{n=1}^{N} X_{ni} + K\alpha} \tag{4}$$

This is the global conditional probability Θ, pooling all N lines. The resulting 3×4 matrix $\Theta = [\theta_{ij}]$ is the core quantity of the EPSPI framework; all eleven estimation methods are either direct estimators of θ_{ij} or functions thereof. Observed values are reported in Table 1.

Table 1: Laplace-Smoothed Θ

	S1: PS	S2: C	S3: CL	S4: CT
EP1: Beh	0.3239	0.0071	0.0360	0.0982
EP2: Cog	0.2138	0.0069	0.0094	0.3591
EP3: Con	0.8061	0.0418	0.8783	0.3118

Constructivism exhibits the highest conditional probabilities for Problem Solving and

Collaborative Learning by a wide margin, reflecting a near-perfect association between Constructivist pedagogy and collaborative skill deployment ($\theta_{\text{Cog,CL}} = 0.0094$ vs. $\theta_{\text{Con,CL}} = 0.8783$). This extreme contrast causes quasi-separation in maximum-likelihood methods. To quantify estimation uncertainty, 95% confidence intervals for $\hat{\theta}_{ij}$ were computed using the Wilson score interval for binomial proportions [1]. The Wilson interval provides better coverage than the normal approximation, particularly for small probabilities.

Let $n_i = \sum_n X_{ni}$ and $n_{ij} = \sum_n X_{ni} Y_{nj}$. The Wilson interval for θ_{ij} with confidence level $1 - \alpha$ is given by

$$CI_{ij} = \frac{\hat{\theta}_{ij} + \frac{z^2}{2n_i} \pm z\sqrt{\frac{\hat{\theta}_{ij}(1-\hat{\theta}_{ij})}{n_i} + \frac{z^2}{4n_i^2}}}{1 + \frac{z^2}{n_i}}, \tag{5}$$

where $z = 1.96$ for a 95% confidence level. Confidence intervals for Behaviourism and

Table 2: Global conditional probability matrix Θ with 95% Wilson confidence intervals. Entries are $P(\text{Skill } j \mid \text{EP}_i)$. (PS: Prob. Solving, C: Creativity, CL: Collab. Learning, CT: Critical Thinking)

	PS	C	CL	CT
Behaviourism	0.324 [0.314, 0.334]	0.007 [0.005, 0.009]	0.036 [0.032, 0.040]	0.098 [0.092, 0.105]
Cognitivism	0.214 [0.207, 0.221]	0.007 [0.006, 0.008]	0.009 [0.008, 0.011]	0.359 [0.351, 0.367]
Constructivism	0.806 [0.763, 0.857]	0.042 [0.021, 0.070]	0.878 [0.844, 0.921]	0.312 [0.259, 0.372]

Cognitivism are relatively narrow due to large sample sizes ($n_i = 8601$ and $n_i = 14422$), while Constructivism exhibits wider intervals ($n_i = 259$), reflecting greater estimation uncertainty.

2.3 Episode-Conditional Probabilities

The global θ_{ij} pools information across all 49 episodes. To capture local pedagogical context within a specific episode, we define the episode-conditional probability:

$$\theta^e_{ij} = \frac{\sum_{\{n:e_n=e\}} X_{ni} Y_{nj} + \alpha}{\sum_{\{n:e_n=e\}} X_{ni} + K\alpha} \tag{6}$$

When EP_i is entirely absent from episode e (denominator $= K\alpha = 4$ before smoothing), the formula returns $\theta_{ij} = \frac{1}{K} = 0.25$, the maximum-entropy prior over four equally probable skills.

2.4 Information-Theoretic Justification

To justify the requirement for a line-level analysis, we employ an information-theoretic framework to quantify the "structural intensity" of the pedagogical patterns and determine

the information loss associated with data aggregation.

2.4.1 Entropy and Mutual Information

We define the configuration pattern of a line n as a 7-bit binary string $z_n \in \{0,1\}^7$, representing the simultaneous state of $P = 3$ Educational Practices and $K = 4$ Skills. The global uncertainty of these patterns is measured by the **Global Entropy** $H(z)$:

$$H(z) = -\sum_{z \in \mathcal{Z}} p(z) \log_2 p(z) \approx 3.140 \text{ bits} \tag{7}$$

To determine the influence of the episode context, we calculate the **Conditional Entropy** $H(z|e)$, which measures the average remaining uncertainty of a pattern given its episode label e:

$$H(z|e) = \sum_{e=1}^{E} \frac{n_e}{N} \left[-\sum_{z \in \mathcal{Z}} p(z|e) \log_2 p(z|e) \right] \approx 1.901 \text{ bits} \tag{8}$$

The **Mutual Information** $I(z, e)$ quantifies the information shared between pedagogical patterns and their episode context:

$$I(z, e) = H(z) - H(z|e) \approx 1.239 \text{ bits} \tag{9}$$

The **Relative Information** ($I/H \approx 39.5\%$) reveals that while episode-level context is significant, 60.5% of the information variance resides at the line-level. This high "intra-episode" variance necessitates the *Three-Layer Architecture* to prevent the loss of fine-grained pedagogical data.

2.4.2 Divergence Metrics: Structural Intensity

To identify specific episodes that deviate from the global pedagogical norm, we define two divergence measures comparing the episode distribution P to the global distribution Q.

2.4.3 Normalized KL Divergence

We use the Kullback–Leibler (KL) divergence to measure the "surprise" of an episode's distribution relative to the corpus, normalized by global entropy to define a **Structural Intensity Index**:

$$D_{KL}(P \parallel Q) = \sum_{z \in \mathcal{Z}} P(z) \log_2 \left(\frac{P(z)}{Q(z)} \right), \quad \text{Norm KL}_e = \frac{D_{KL}(P \parallel Q)}{H(z)} \tag{10}$$

2.4.4 Jensen–Shannon Divergence (JSD)

To provide a symmetric and bounded distance metric ($0 \leq JSD \leq 1$), we utilize the Jensen–Shannon Divergence, which compares both distributions to an average $M = \frac{1}{2}(P + Q)$:

$$JSD(P \parallel Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M) \tag{11}$$

2.4.5 Permutation Test for Statistical Significance

To ensure the observed $I(z, e)$ is not a result of the large number of episodes ($E = 49$), a permutation test was conducted ($2,000$ iterations). Under the null hypothesis H_0 (no relationship between pattern and episode), the episode labels were randomly shuffled.

- **Observed MI:** 1.239 bits
- **Null Distribution:** $\mu = 0.053, \sigma = 0.0015$
- **p-value:** < 0.0005

The observed MI is significantly higher (> 700 standard deviations) than the null mean, confirming that the pedagogical structures are highly episode-dependent and non-random.

Table 3: Entropy and Information Results

Metric	Value
Global Entropy $H(z)$	3.140 bits
Conditional Entropy $H(z\|e)$	1.901 bits
Mutual Information $I(z, e)$	1.239 bits
Relative Information Explained	39.47%

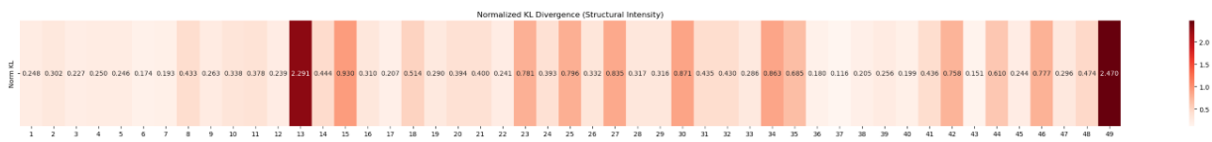

Figure 2: Normalized KL Divergence (Structural Intensity)

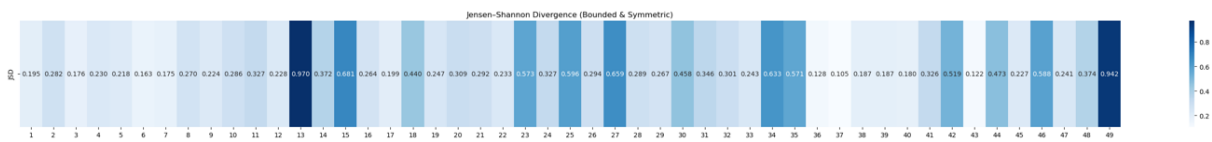

Figure 3: Jensen–Shannon Divergence (Bounded & Symmetric)

2.5 Three-Layer EPSPI Architecture

The EPSPI framework is constructed in three layers to preserve within-episode information before aggregation.

Layer 1: Line-level EPSPI

$$\mathrm{EPSPI}_n = f(X_{n\cdot}, \Theta, \hat{\theta}^{e_n}), \quad n = 1, \ldots, N \tag{12}$$

Layer 2: Episode aggregation

$$\mathrm{EPSPI}_e = \frac{1}{n_e} \sum_{\{n:e_n=e\}} \mathrm{EPSPI}_n \tag{13}$$

Layer 3: Validation target The observed skill rate for episode e, computed independently from all EP and EPSPI quantities, is:

$$\bar{Y}_e = \frac{1}{n_e} \sum_{\{n:e_n=e\}} \left(\frac{1}{K} \sum_{j=1}^{K} Y_{nj} \right) \tag{14}$$

The primary validation criterion is the cross-validated correlation CV $r(\mathrm{EPSPI}_e, \bar{Y}_e)$. The decomposition $\bar{Y}_e = \mathrm{EPSPI}_e + \varepsilon_e$ defines the non-EP residual ε_e.

3 Evaluation Framework

3.1 Episode-Level Metrics

All primary metrics are computed at the episode level ($E = 49$) to respect the hierarchical structure of the data.

3.1.1 Train Pearson r.

$$r_{\text{train}} = \mathrm{corr}\big(\mathrm{EPSPI}_e,\, \bar{Y}_e\big), \quad e = 1, \ldots, 49. \tag{15}$$

3.1.2 Calibrated Explained Variance (Exp.Var(cal)).

[2, 3] An OLS calibration $\hat{Y}_e = a + b \cdot \mathrm{EPSPI}_e$ is fitted; explained variance is then:

$$\mathrm{Exp.Var(cal)} = 1 - \frac{\mathrm{Var}(\bar{Y}_e - \hat{Y}_e^{\text{cal}})}{\mathrm{Var}(\bar{Y}_e)}. \tag{16}$$

Calibration corrects for scale differences between EPSPI and $\bar{Y}_e$.

3.1.3 Raw Explained Variance (Exp.Var(raw)).

[2, 3]

$$\text{Exp.Var(raw)} = 1 - \frac{\text{Var}(\bar{Y}_e - \text{EPSPI}_e)}{\text{Var}(\bar{Y}_e)}. \tag{17}$$

Negative values indicate the raw EPSPI scale is offset from $\bar{Y}_e$; the gap Exp.Var(cal) − Exp.Var(raw) quantifies scale bias per method.

3.1.4 Cross-Validated Pearson r.

Five-fold CV with GroupKFold splits at the episode level (no line from a test episode appears in the training fold):

$$\text{CV}\, r = \frac{1}{5}\sum_{k=1}^{5} \text{corr}\big(\text{EPSPI}_e^{(k)},\, \bar{Y}_e\big), \quad e \in \text{test fold } k. \tag{18}$$

3.1.5 Cross-Validated MSE.

$$\text{CV MSE} = \frac{1}{5}\sum_{k=1}^{5} \frac{1}{|F_k|} \sum_{e \in F_k} \big(\text{EPSPI}_e - \bar{Y}_e\big)^2. \tag{19}$$

3.2 Line-Level Metrics

3.2.1 AIC and BIC.

For each method, a fixed external logistic model is fitted at the line level:

$$\text{logit}\big(P(Y_{nj} = 1)\big) = \beta_{0j} + \beta_{1j} \cdot \text{EPSPI}_n, \quad j = 1, \ldots, 4. \tag{20}$$

AIC and BIC are computed from the log-likelihood of this model, summed across all four skills:

$$\text{AIC} = \sum_{j=1}^{4} \big(2k_j - 2\ell_j\big), \quad \text{BIC} = \sum_{j=1}^{4} \big(k_j \ln N - 2\ell_j\big), \tag{21}$$

where $k_j = 2$ (intercept + slope) and ℓ_j is the maximised log-likelihood. Because k_j and N are identical across all methods, AIC/BIC differences reflect log-likelihood differences exclusively.

Remark 1. AIC and BIC are meaningful here even for non-parametric methods (RF, SVM, NMF) because the likelihood is computed from the fixed external logistic model, not from the EPSPI method itself. This provides a consistent line-level discriminative comparison across all eleven methods.

3.3 Rationale and Worked Examples for Each Metric

The six metrics collectively address three distinct questions about an EPSPI method: (1) does it *rank* episodes correctly? (2) does it explain episode-level *variance* in skill deployment? and (3) does it *discriminate* skill presence at the line level? Each question is necessary but not sufficient alone. A method could rank episodes perfectly (high r) but with a wildly wrong scale (negative Exp.Var(raw)); conversely, a method could fit the scale correctly in-sample but collapse under CV. The six metrics are therefore complementary, not redundant.

3.3.1 Train Pearson r: Rank-Order Agreement

What it measures. Pearson r measures the linear association between predicted EPSPI_e and observed $\bar{Y}_e$ across all 49 episodes. A value of $r = 1$ means the method perfectly ranks episodes by their skill deployment; $r = 0$ means EP structure carries no signal.

Why use it. Pearson r is the standard measure of predictive validity in educational measurement [4] and has been used as the primary metric in studies relating pedagogical features to learning outcomes [5, 6]. It is interpretable: r^2 is the proportion of variance explained (equivalent to R^2 in simple linear regression). In our context it answers the pedagogical question directly: do lessons coded with more skill-conducive EPs actually produce more skill-present moments?

Worked example. Suppose five episodes have the following values (a subset of $E = 49$):

Episode	EPSPI_e (predicted)	$\bar{Y}_e$ (observed)
Ep 3	0.12	0.15
Ep 7	0.16	0.19
Ep 13	0.51	0.61
Ep 21	0.14	0.11
Ep 38	0.18	0.22

The rankings are almost identical (Ep13 highest in both, Ep3/21 lowest), giving a high r. Now suppose a broken method predicts $\text{EPSPI}_e = 0.15$ for every episode regardless of EP structure. Then $r = 0$ because a constant prediction has zero variance and pearson r is undefined (or zero by convention). This is exactly the failure mode of methods with low $P(\text{EP}_{\text{Con}})$ signal.

Limitation. Train r is computed on all 49 episodes including those used to estimate θ_{ij}. It therefore measures in-sample fit, not predictive validity on new episodes. CV r (below) corrects for this.

3.3.2 Exp.Var(cal) vs Exp.Var(raw): Scale vs Rank

What they measure. Both measure how much of the variance in $\bar{Y}_e$ is explained, but they differ in whether scale is penalised.

Exp.Var(raw) uses the raw EPSPI predictions without any rescaling:

$$\text{Exp.Var(raw)} = 1 - \frac{\text{Var}(\bar{Y}_e - \text{EPSPI}_e)}{\text{Var}(\bar{Y}_e)}.$$

If EPSPI_e is systematically too low or too high (scale bias), the residuals $\bar{Y}_e - \text{EPSPI}_e$ are large even if the ranking is correct, and Exp.Var(raw) can be negative.

Exp.Var(cal) first fits an OLS calibration $\hat{Y}_e = a + b \cdot \text{EPSPI}_e$ and computes:

$$\text{Exp.Var(cal)} = 1 - \frac{\text{Var}(\bar{Y}_e - \hat{Y}_e^{\text{cal}})}{\text{Var}(\bar{Y}_e)}.$$

This is numerically identical to R^2 from the OLS regression of $\bar{Y}_e$ on EPSPI_e, which equals r^2 for simple regression. Calibration removes scale bias before computing variance explained.

Why use both. The *gap* Exp.Var(cal) − Exp.Var(raw) is a direct diagnostic of scale bias [7]. Methods F (PCA) and H (NMF) produce $\text{EPSPI}_e \approx 0.03$–$0.05$ because the principal component or NMF factor is anchored to the Constructivism direction, compressing most episodes near zero. The raw predictions are so far below $\bar{Y}_e \approx 0.20$ that Exp.Var(raw) is *negative* (the method is worse than predicting the mean), yet Exp.Var(cal) is positive (≈ 0.49) because the *ranking* is correct. Reporting only Exp.Var(cal) would hide this scale failure; reporting only Exp.Var(raw) would penalise a method that is conceptually sound but needs rescaling for deployment. Both together give the complete picture.

Worked example. Using the same five episodes with Method F (PCA):

Ep	$\bar{Y}_e$	EPSPI_e^C (raw)	$\hat{Y}_e^{\text{cal}}$	Raw residual
Ep 3	0.150	0.021	0.148	−0.129
Ep 7	0.190	0.038	0.191	−0.152
Ep 13	0.610	0.180	0.608	−0.430
Ep 21	0.110	0.015	0.112	−0.095
Ep 38	0.220	0.044	0.218	−0.176

The raw residuals are large (method predicts ∼0.02-0.18 when true values are 0.11-0.61), giving negative Exp.Var(raw). But the *calibrated* predictions $\hat{Y}_e^{\text{cal}}$ track $\bar{Y}_e$ closely, giving high Exp.Var(cal). The OLS slope $b \approx 3.2$ rescales PCA predictions into the correct range — the information is there, just on the wrong scale.

3.3.3 CV Pearson r: Out-of-Sample Predictive Validity

What it measures. CV r is the mean Pearson correlation between predicted and observed $\bar{Y}_e$ on *held-out* episodes across five folds. It estimates how well an EPSPI method would predict skill deployment in a new episode never seen during fitting.

Why use it. In-sample metrics (Train r, Exp.Var) can be inflated by overfitting, especially for methods with many parameters (RF, SVM, GEE with separate coefficients per skill). CV r is the gold standard for predictive validity in educational data mining [8, 9]. The episode-grouped split is essential: if lines from the same episode appeared in both train and test folds, the θ^e_{ij} matrix would leak ground-truth skill rates into the features, inflating CV r artificially [10, 11]. GroupKFold at the episode level guarantees no such leakage.

Worked example: what a fold looks like. With $E = 49$ and 5 folds, each test fold contains approximately 10 episodes ($\approx$ 3,836 lines on average).

- **Fold 2 test:** Episodes {4, 9, 14, 20, 27, 33, 38, 44, 48, 49}
- **Fold 2 train:** all remaining 39 episodes

The method (e.g. GEE) is re-estimated from scratch on the 39 training episodes. New θ^e_{ij} matrices are computed for the 10 test episodes using only those episodes' own lines (no training data leaks in). EPSPI_e is computed for each test episode, compared to its $\bar{Y}_e$, and Pearson r is recorded. This repeats for all 5 folds; the mean is CV r.

Relationship to Train r. The drop from Train $r \approx 0.70$ to CV $r \approx 0.45$ is expected and healthy. A drop of 0.25 is not evidence of overfitting — it reflects the genuine loss of in-sample θ^e_{ij} calibration when predicting new episodes. If CV $r \approx$ Train r, the method has very low variance (e.g. Method A, which uses no learned parameters beyond θ). If CV $r \ll$ Train r (e.g. SVM: $0.85 \to 0.31$), the method overfits the 49-episode training set.

3.3.4 CV MSE: Magnitude of Prediction Error

What it measures. CV MSE measures the average squared prediction error per episode on held-out folds:

$$\text{CV MSE} = \frac{1}{5}\sum_{k=1}^{5}\frac{1}{|F_k|}\sum_{e\in F_k}\left(\text{EPSPI}_e - \bar{Y}_e\right)^2.$$

Unlike CV r, which is scale-invariant, MSE penalises both rank errors and scale errors simultaneously.

Why use it alongside CV r. CV r can be high while MSE is also high if predictions are monotonically correct but far from the true values in magnitude. MSE is the standard loss function for regression tasks [12] and directly measures operational usefulness: a teacher or curriculum designer who uses EPSPI_e as an absolute estimate of expected skill rate needs low MSE, not just high rank-order correlation.

Worked example. Method C (Logistic) achieves CV $r = 0.400$ and CV MSE $= 0.008525$. Method F (PCA) achieves CV $r = 0.281$ but CV MSE $= 0.042322$ — five times larger. PCA's scale compression (predictions near 0.04 when true values are $\sim$0.20) inflates MSE even though the ranking is partially preserved. The GEE method achieves the best CV $r = 0.450$ *and* the lowest CV MSE $= 0.008018$, confirming it as the overall best method.

3.3.5 AIC and BIC: Line-Level Discriminative Power

What they measure. For each method, a fixed external logistic regression is fitted at the line level:

$$\text{logit}\big(P(Y_{nj} = 1)\big) = \beta_{0j} + \beta_{1j} \cdot \text{EPSPI}_n, \quad j = 1, \ldots, 4.$$

AIC and BIC are computed from the summed log-likelihood of this model across all four skills:

$$\text{AIC} = \sum_{j=1}^{4} (2k_j - 2\ell_j), \qquad \text{BIC} = \sum_{j=1}^{4} (k_j \ln N - 2\ell_j),$$

where $k_j = 2$ (intercept + slope) and $N = 19{,}181$. Lower AIC/BIC means the logistic model fits better with that method's EPSPI_n as the predictor.

Why use AIC/BIC for non-parametric methods. AIC [13] and BIC [14] are normally used to compare parametric models of different complexity. Here they are applied consistently across all 11 methods — including non-parametric ones (RF, SVM, NMF) — because the *likelihood comes from the fixed external logistic model*, not from the EPSPI method itself. Since $k_j = 2$ and $N = 19{,}181$ are identical for all methods, AIC differences are log-likelihood differences exclusively. This provides a principled line-level comparison without assuming the methods are parametric [15].

Worked example: what AIC differences mean. The Correspondence Analysis (CA) method achieves the lowest AIC (44,358), while the Random Forest achieves the highest (48,508). The difference $\Delta\text{AIC} = 4{,}150$ is decisive on the [15] scale (differences > 10 are considered very strong evidence against the higher-AIC model). CA wins AIC because it directly optimises the chi-squared EP–skill co-occurrence structure at the line level — exactly what the logistic likelihood rewards. However, CA achieves only CV

$r = 0.296$ at the episode level, revealing a tension: line-level discriminative power (AIC) and episode-level predictive validity (CV r) measure different aspects of performance. A method can be excellent at one and poor at the other. Reporting both together exposes this tension transparently.

Numerical guide. On the [15] scale: ΔAIC < 2 = essentially equivalent; 2–7 = moderate evidence for lower-AIC method; > 10 = decisive. In our results, all pairwise ΔAIC > 100, meaning the differences are unambiguous.

3.4 Why Six Metrics and Not One

No single metric captures the full picture [16, 17]:

Metric	Question answered	Blind spot
Train r	Rank agreement in-sample	Scale; overfitting
Exp.Var(cal)	Rank + variance in-sample	Scale bias hidden
Exp.Var(raw)	Absolute scale agreement	Penalises good rankers
CV r	Out-of-sample ranking	Scale; magnitude
CV MSE	Out-of-sample magnitude	Rank order
AIC/BIC	Line-level discrimination	Episode-level aggregation

The GEE method is the recommended choice because it achieves the best CV r (0.450) and the lowest CV MSE (0.008018) simultaneously. AIC and Exp.Var(raw) are reported for completeness and to make the scale-bias pattern transparent.

4 Eleven Estimation Methods

We organise the eleven frequentist formulations of EPSPI along three axes of statistical inquiry, each reflecting a distinct scientific question about the EP–skill relationship (Table 4). The first axis — **direct estimation** — asks what value $\hat{\theta}_{ij}$ takes under progressively stronger modelling assumptions [18]. Methods A and B establish a non-parametric baseline using raw conditional frequencies [19], differing only in whether EP weights are equal or frequency-proportional. Method C introduces the first parametric assumption — that the log-odds of skill presence is linear in EP indicators — via logistic regression [20], and Method D extends this with a within-episode clustering correction through GLMM random intercepts [21, 22]. Method GEE completes this axis by relaxing the within-episode independence assumption without specifying a full likelihood, estimating population-average EP–skill associations robust to within-episode correlation [23, 24]. The second axis — **Dimensionality reduction** — asks whether the $P \times K = 12$-dimensional EP–skill profile space contains latent structure that improves estimation of $\hat{\theta}_{ij}$

[25]. Method G (PCA) finds linear components of maximum variance without constraints [26], Method CA (Correspondence Analysis) discovers the principal axes of EP–skill co-occurrence in a contingency-table framework [27], and Method H (NMF) adds the domain-appropriate non-negativity constraint, discovering additive teaching-mode profiles that reconstruct each episode's EP–skill fingerprint [28]. The third axis — **predictive modelling** — asks how accurately skill expression can be predicted from EP patterns using methods that differ in their assumptions about the data-generating process [29]. Method I (Naïve Bayes) inverts the conditional probability via Bayes' theorem under a conditional-independence assumption [30], Method RF (Random Forest) learns non-linear EP–skill interactions through an ensemble of decorrelated trees [31], and Method K (SVM) abandons distributional assumptions entirely, seeking instead the maximum-margin boundary in a kernel-induced feature space [32, 33].

Together, the eleven methods span the major schools of frequentist statistical thought — non-parametric, parametric, hierarchical, robust, latent-variable, generative, and discriminative — applied to a common estimand: the EP–skill affinity matrix $\hat{\theta}_{ij} = \hat{P}(S_j \mid \mathrm{EP}_i)$. This breadth serves two purposes: it demonstrates that substantive conclusions about EP–skill relationships are robust across methodological choices [34], and it reveals where and why methods diverge, illuminating the structure of the EPSPI estimation problem itself.

Table 4: Organisation of the eleven EPSPI estimation methods along three axes of statistical inquiry.

Axis	**Methods**	**Characteristic**
Direct Estimation	A, B, C, D, E	Model $P(S_j \mid \mathrm{EP}_i)$ explicitly under progressively stronger parametric assumptions
Dimensionality reduction	F, G, H	Reduce the $P \times K$ EP–skill space to latent dimensions; $\hat{\theta}_{ij}$ is recovered from low-rank structure
Predictive Modelling	I, J, K	Treat EPSPI_n as a regression target; discriminative models trained to minimise prediction error

Methods are organised along three axes:

4.1 Method A: Simple Average

Method A is the direct plug-in estimator of EPSPI under equal weighting of all active EPs and all skills. It is a **non-parametric estimator**: the only quantities entering the formula are the Laplace-smoothed global conditional probabilities θ_{ij} (Equation (**??**)), computed once from all N lines and fixed thereafter. No parameters are fitted to episode-level

outcomes.

Line-level score. Let $p_n = \sum_{i=1}^{P} X_{ni}$ be the number of EPs active in line n. The Method A score for line n is:

$$\mathrm{EPSPI}_n^{(A)} = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni} \left(\frac{1}{K} \sum_{j=1}^{K} \theta_{ij} \right), \qquad p_n > 0, \tag{22}$$

where θ_{ij} is the **global** Laplace-smoothed conditional probability matrix (Equation (**??**)). The indicator X_{ni} selects only the active EPs; dividing by p_n ensures a line with one active EP and a line with three active EPs contribute equally to the episode mean. When $p_n = 0$ — which does not occur in this corpus by annotation design — the fallback $\mathrm{EPSPI}_n^{(A)} = (PK)^{-1} \sum_{i=1}^{P} \sum_{j=1}^{K} \theta_{ij}$ is used, which is the grand mean of Θ.

Deployment example. For a line with Cognitivism and Constructivism active ($X_{n1} = 0$, $X_{n2} = 1$, $X_{n3} = 1$, so $p_n = 2$):

$$\mathrm{EPSPI}_n^{(A)} = \frac{1}{2}[\frac{1}{4}(0.429+0.143+0.286+0.429)+\frac{1}{4}(0.429+0.143+0.571+0.286)] = 0.3395.$$

A practitioner needs only the published Θ table (Table 2) and their own EP binary flags X_{ni} — no skill observations Y_{nj} are required.

Episode-level aggregation. The episode-level score is the unweighted mean over the n_e lines belonging to episode e:

$$\mathrm{EPSPI}_e^{(A)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n=e\}} \mathrm{EPSPI}_n^{(A)}, \tag{23}$$

where $n_e = |\{n : e_n = e\}|$ is the actual number of lines in episode e. No equal-size assumption is imposed across episodes.

Why global θ_{ij} and not episode-conditional $\theta_{ij}^{e_n}$. Method A deliberately uses the global θ_{ij} for two reasons. First, $\theta_{ij}^{e_n}$ is estimated from the skill observations Y_{nj} of the *same* episode used to compute the validation target $\bar{Y}_e$, introducing a direct within-episode dependency between predictor and outcome that the cross-validated evaluation is designed to prevent. Second, Method A is a **non-parametric plug-in estimator**: it substitutes observed conditional frequencies directly into the formula without fitting any functional form to Y_{nj}. Methods C_Logistics, D_GLMM, and E_GEE also use episode-conditional information but do so through a parametric likelihood fitted jointly across all episodes — their parameters $\hat{\beta}$ are estimated from training episodes and applied to test episodes where Y_{nj} is never observed during fitting, avoiding the same circularity. The CV gap

between Methods A/B (global θ) and C/D/E (episode-conditional estimation) therefore quantifies the predictive gain from incorporating within-episode skill information under proper cross-validation.

Properties. Method A has **zero free parameters**. Because p_n weights each active EP equally regardless of its corpus frequency, Method A does not distinguish an episode dominated by a single rare EP (e.g. pure Constructivism, $\delta^e_{\text{Con}} = 1.00$) from one where that EP co-occurs with others at equal frequency — a limitation addressed by Method B, which replaces the line-level X_{ni} weights with episode-level proportions δ^e_i.

4.2 Method B: δ-Weighted Average

Method A weights each active EP by its line-level activation X_{ni}, normalised by p_n. Method B replaces this with the **episode EP proportion** δ^e_i — how dominant each pedagogical paradigm is across the whole episode. A line embedded in a predominantly Constructivist episode receives a higher Constructivism weight than the same line in a mixed episode, even if its own EP flags are identical.

Episode EP proportions δ^e_i. For each episode e and EP i:

$$\delta^e_i = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} X_{ni}, \tag{24}$$

where n_e is the number of lines in episode e. $\delta^e_i \in [0, 1]$ is the proportion of lines in which EP_i is active. Because multiple EPs may be simultaneously active in the same line, $\sum_{i=1}^{P} \delta^e_i$ can exceed 1. δ^e_i requires only X_{ni} — no skill observations Y_{nj} — so it can be computed before or without skill coding, carrying no leakage at deployment.

Example: Episode 7. Using the six-line episode from Section **??** (Table **??**), recall the EP annotation counts:

	Beh ($i = 1$)	Cog ($i = 2$)	Con ($i = 3$)
$\sum_{\{n:e_n=7\}} X_{ni}$	3	3	3
$n_e = 6$			
$\delta^{e=7}_i$	$3/6 = 0.500$	$3/6 = 0.500$	$3/6 = 0.500$

In this balanced episode all three EPs appear equally often, so $\delta^{e=7}_i = 0.5$ for all i. In a real episode such as Episode 13 (pure Constructivism), $\delta^{13}_{\text{Con}} = 1.00$ and $\delta^{13}_{\text{Beh}} = \delta^{13}_{\text{Cog}} = 0.00$, which is why Episode 13 is a high-leverage point — its δ profile is an extreme vertex of the simplex.

Line-level score. The Method B score for line n uses $\delta_i^{e_n}$ as the EP weight and the row mean of Θ as the skill summary:

$$\mathrm{EPSPI}_n^{(B)} = \sum_{i=1}^{P} \delta_i^{e_n} \left(\frac{1}{K} \sum_{j=1}^{K} \theta_{ij} \right), \tag{25}$$

where θ_{ij} is the global Laplace-smoothed conditional probability (Equation (**??**)). The inner average $\frac{1}{K}\sum_j \theta_{ij}$ is the i-th row mean of Θ — no new symbol is required.

Example: Episode 7. Using $\delta_i^{e=7} = 0.5$ for all i and the global Θ matrix (Table 1):

$$\frac{1}{K}\sum_j \theta_{\mathrm{Beh},j} = \tfrac{1}{4}(0.3239 + 0.0071 + 0.0360 + 0.0982) = 0.116$$
$$\frac{1}{K}\sum_j \theta_{\mathrm{Cog},j} = \tfrac{1}{4}(0.2138 + 0.0069 + 0.0094 + 0.3591) = 0.147$$
$$\frac{1}{K}\sum_j \theta_{\mathrm{Con},j} = \tfrac{1}{4}(0.8061 + 0.0418 + 0.8783 + 0.3118) = 0.510$$

Every line in Episode 7 receives the same score:

$$\mathrm{EPSPI}_n^{(B)} = 0.5 \times 0.116 + 0.5 \times 0.147 + 0.5 \times 0.510 = 0.058 + 0.074 + 0.255 = \mathbf{0.387}$$

Compare this to the Method A scores for the same episode, which varied from 0.286 (Beh-only line) to 0.357 (Con-only line): Method B replaces that within-episode variation with a single episode-constant value, trading granularity for deployability.

Episode-level aggregation. Because $\mathrm{EPSPI}_n^{(B)}$ is identical for all n_e lines within episode e, the episode mean collapses to:

$$\mathrm{EPSPI}_e^{(B)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(B)} = \sum_{i=1}^{P} \delta_i^{e} \left(\frac{1}{K} \sum_{j=1}^{K} \theta_{ij} \right), \tag{26}$$

which is the dot product of the episode EP profile $[\delta_1^e, \delta_2^e, \delta_3^e]$ with the fixed row-mean vector of Θ.

What the evaluation metrics mean for Methods A and B. Methods A and B have **zero free parameters**: θ_{ij} is computed directly from frequency counts and fixed before any evaluation. Because no parameters are fitted to the 49 episodes, the distinction between "training" and "testing" does not apply in the conventional sense. The six evaluation metrics therefore have the following interpretations for these methods:

- **In-sample Pearson** r (labelled Train r in the results table) is the Pearson correlation between $\text{EPSPI}_e^{(A/B)}$ and $\bar{Y}_e$ computed over all 49 episodes. There is no overfitting risk because no episode-level parameters were estimated; the label "train" is a misnomer and should be read as "in-sample association."

- **Exp.Var(cal)** is the R^2 from an OLS regression of $\bar{Y}_e$ on $\text{EPSPI}_e^{(A/B)}$ over all 49 episodes. The OLS calibration ($\hat{Y}_e = a + b \cdot \text{EPSPI}_e$) uses 2 parameters but these are not part of the EPSPI estimator itself — they are a post-hoc rescaling used only for evaluation.

- **Exp.Var(raw)** measures the same thing without the OLS rescaling, penalising scale mismatch directly.

- **CV Pearson** r is the only metric where the zero-parameter nature of A and B genuinely matters. In each fold, θ_{ij} is recomputed from the 39 training episodes only, then frozen; $\text{EPSPI}_e^{(A/B)}$ is computed for the 10 test episodes using this held-out θ_{ij} estimate. This measures whether the EP–skill affinity estimated from 39 episodes generalises to 10 unseen ones. For zero-parameter methods, the expected drop from in-sample r to CV r is small — only the variance in θ_{ij} estimation across folds contributes, not model overfitting.

- **CV MSE** follows the same fold structure as CV r and measures the magnitude of prediction error on held-out episodes.

- **AIC/BIC** are computed from a fixed external logistic model at the line level ($\text{logit}\, P(Y_{nj} = 1) = \beta_{0j} + \beta_{1j} \cdot \text{EPSPI}_n^{(A/B)}$, $k_j = 2$). This model has its own 2 parameters per skill (fitted at the line level), but these are the evaluator's parameters, not Method A/B's. AIC/BIC therefore measure how well $\text{EPSPI}_n^{(A/B)}$ discriminates skill presence at the line level.

For methods with free parameters (E, G, I, J, K), the in-sample metrics reflect genuine overfitting risk and the Train/CV gap is substantively meaningful. For Methods A and B, the Train/CV gap reflects only estimation variance in θ_{ij} across the 5 CV folds.

Properties. Method B has zero free parameters and is fully deployable from episode-level EP records alone. Its key limitation is that $\text{EPSPI}_n^{(B)}$ is episode-constant: all lines within episode e receive the same score, so within-episode EP dynamics — a lesson that shifts from Behaviourism to Constructivism mid-episode — are invisible to Method B but captured by Method A's line-varying X_{ni}/p_n weights.

4.3 Method C: Logistic Regression

Methods A and B estimate $P(S_j \mid \mathrm{EP}_i)$ as a raw conditional frequency — the proportion of lines with both EP i and skill j active. Method E replaces this non-parametric plug-in with a **parametric logistic model**, assuming that the log-odds of skill S_j being present is a linear function of the EP binary indicators. This introduces the first functional-form assumption in the EPSPI framework, at the cost of $K(P+1) = 16$ free parameters.

Model specification. For each skill $j \in \{1, \ldots, K\}$, a binary logistic regression is fitted at the line level:

$$\operatorname{logit} \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) = \beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni}, \tag{27}$$

where β_{0j} is the skill-specific intercept and β_{ij} is the log-odds change in skill S_j associated with EP_i being active. The model is fitted separately for each of the $K = 4$ skills, yielding a $P \times K$ coefficient matrix $B = [\beta_{ij}]$ and a K-vector of intercepts $[\beta_{0j}]$. Estimation uses maximum likelihood over all $N = 19{,}181$ lines.

The predicted probability of skill S_j for line n is:

$$\hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) = \sigma\left(\beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni}\right), \tag{28}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic function.

Relationship to θ_{ij}. In Method A, the conditional probability of skill S_j given only EP_i active ($p_n = 1$, $X_{ni} = 1$, $X_{ni'} = 0$ for $i' \neq i$) is directly θ_{ij}. In Method C, the same quantity is:

$$\hat{P}(Y_{nj} = 1 \mid X_{ni} = 1,\, X_{ni'} = 0\ \forall i' \neq i) = \sigma(\beta_{0j} + \beta_{ij}). \tag{29}$$

Method E therefore implicitly defines a parametric analogue of θ_{ij}:

$$\tilde{\theta}_{ij}^{(E)} = \sigma(\beta_{0j} + \beta_{ij}),$$

constrained to follow the logistic functional form, unlike the unconstrained frequency estimate θ_{ij} of Method A. When EP co-occurrence is present ($p_n > 1$), Method E captures the joint effect through the additive log-odds structure, whereas Method A averages θ_{ij} values independently.

Line-level score. The Method E score for line n is the mean predicted skill probability across all K skills:

$$\mathrm{EPSPI}_n^{(E)} = \frac{1}{K} \sum_{j=1}^{K} \sigma\left(\beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni}\right). \tag{30}$$

The full EP flag vector $X_{n\cdot}$ enters the formula — not just the active EPs — because the logistic model predicts skill probability for any combination of EP flags simultaneously. The p_n normalisation of Method A is therefore not needed: multi-EP lines are handled naturally through the additive log-odds.

Deployment example. For a line with only Constructivism active ($X_{n1} = 0$, $X_{n2} = 0$, $X_{n3} = 1$):

$$\mathrm{EPSPI}_n^{(E)} = \frac{1}{4} \sum_{j=1}^{4} \sigma(\beta_{0j} + \beta_{3j}).$$

A practitioner needs only the published coefficient vectors $[\beta_{0j}, \beta_{1j}, \beta_{2j}, \beta_{3j}]$ for each skill j and their own EP binary flags X_{ni} — the same input requirement as Method A, but the published parameters are regression coefficients rather than a frequency table.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(E)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(E)}, \tag{31}$$

where n_e is the actual number of lines in episode e.

What the evaluation metrics mean for Method E. Unlike Methods A and B, Method E has $K(P+1) = 4 \times 4 = 16$ free parameters fitted to Y_{nj} via maximum likelihood. The evaluation metrics therefore carry a different interpretation:

- **In-sample Pearson r and Exp.Var(cal)** reflect genuine in-sample fit of a parametric model. Overfitting risk is small — 16 parameters on 19,181 lines is a highly over-determined system — but nonzero. The label "Train r" is accurate here, unlike Methods A/B where it was a misnomer for in-sample association.
- **CV Pearson r and CV MSE** are the primary reliability metrics. In each fold the 16 coefficients are re-estimated from ∼15,000 training lines and applied to ∼4,000 test lines whose Y_{nj} was never seen during fitting. The Train/CV gap for Method E reflects both estimation variance and genuine model overfitting — in contrast to Methods A/B where only θ_{ij} estimation variance contributes.
- **AIC/BIC** use $k = K(P+1) = 16$ in the penalty, versus $k = 0$ for Methods A and B. Lower AIC despite the heavier penalty indicates genuine log-likelihood improvement from the logistic functional form over raw frequency estimation.

Quasi-separation and Constructivism. Constructivism is active in only 1.1% of lines ($\sum_n X_{n,\mathrm{Con}} \approx 211$) yet has the highest skill co-occurrence rates in Θ ($\theta_{\mathrm{Con,PS}} = 0.806$,

$\theta_{\text{Con,CL}} = 0.878$). This extreme imbalance causes **quasi-complete separation** in the logistic model for Problem Solving and Collaborative Learning: the 211 Constructivist lines nearly perfectly predict high skill presence, driving $\hat{\beta}_{3j} \to \infty$. This is handled by L2 (ridge) regularisation with $\lambda = 1.0$, which shrinks the Constructivism coefficients toward zero and produces finite, stable estimates at the cost of slight underestimation of $\hat{P}(S_j = 1 \mid \text{Con})$ relative to the raw frequency θ_{ij}. Method G (GLMM) provides a robustness check by fitting the same logistic structure with random episode intercepts, testing whether the fixed-effect estimates of Method E are distorted by within-episode correlation.

Properties. Method E has $K(P+1) = 16$ free parameters and is the first parametric method in the framework. Its primary limitation is the additive log-odds assumption: the effect of EP_i on $\text{logit}\, P(S_j = 1)$ is assumed independent of which other EPs are simultaneously active, which may not hold when EP co-occurrence is pedagogically meaningful. Method G relaxes the within-episode independence assumption by adding random intercepts; Method I (GEE) retains the linear log-odds structure but replaces the likelihood with a robust estimating equation, removing the need to specify the within-episode correlation structure entirely. This section details the mathematical formulation and optimization strategy for **Method C (Logistic Regression)**, focusing on how the framework handles the "quasi-separation" observed in classroom pedagogical data through regularization.

Logistic Regression and Regularization

This section details the mathematical formulation and optimisation strategy for Method C, focusing on how the framework handles the "quasi-separation" observed in classroom pedagogical data through L_2 regularisation.

In this corpus, certain EPs—specifically Constructivism—exhibit a "near-perfect" association with skills like Collaborative Learning ($\theta_{Con,CL} = 0.8783$). In a standard maximum-likelihood estimation, this "quasi-separation" causes the weights β to diverge toward infinity, leading to numerical instability and overfitted "all-or-nothing" predictions. This happens because in the sigmoid function, to get an output near 1, the coefficient of one of the features should be very high so that the expression $e^{-\beta X}$ goes near zero, and the sigmoid function can give an output of 1. The L_2 regularization addresses this by penalizing the squared magnitude of the coefficients. This "shrinkage" forces the model to find a finite equilibrium where the marginal gain in log-likelihood from a larger β is offset by the increasing penalty cost. This ensures the line-level $EPSPI_n$ remains a nuanced probability rather than a binary output

The Regularized Objective Function

Method C is a discriminative model that directly estimates the probability of a skill j being present given the active Educational Practices (EPs) on a line n. To ensure stability across the corpus of $N = 19,181$ lines, we minimize the negative log-likelihood (Log-Loss) with an added L_2 (Ridge) penalty:

$$J(\beta_j) = -\frac{1}{N}\sum_{n=1}^{N}[Y_{nj}\ln(\hat{p}_{nj}) + (1 - Y_{nj})\ln(1 - \hat{p}_{nj})] + \frac{1}{2C}\sum_{i=1}^{P}\beta_{ij}^2 \tag{32}$$

Where:

- $\hat{p}_{nj}$ is the predicted probability for skill j on line n:

$$\hat{p}_{nj} = \sigma\left(\beta_{0j} + \sum_{i=1}^{P}\beta_{ij}X_{ni}\right) \tag{33}$$

- $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid activation function.
- C is the inverse regularization strength (set to 1.0).
- $\frac{1}{2C}\sum\beta_{ij}^2$ is the L_2 penalty term that constrains the magnitude of the coefficients.

Optimization via L-BFGS

To solve for the optimal β vector, the framework utilizes the **Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS)** algorithm. Unlike standard Gradient Descent, which only utilizes first-order derivatives (the gradient), L-BFGS is a quasi-Newton method that approximates the **Hessian matrix** (the second-order curvature of the loss function). Analogy: If Gradient Descent is like walking down a mountain in the fog looking only at your feet, L-BFGS is like having a map that tells you how the slope changes further down.

- **Curvature Awareness:** By estimating the curvature, L-BFGS takes more informed steps toward the global minimum, avoiding the numerical oscillations common in Gradient Descent when navigating the "narrow valleys" characteristic of correlated educational data.
- **Memory Efficiency:** It stores a limited history of the most recent gradients to approximate the inverse Hessian, making it computationally efficient for the high-dimensional pattern space ($z_n \in \{0,1\}^7$) of the EPSPI framework.
- **Convergence:** The model is configured with a maximum of $2,000$ iterations to ensure convergence even in the presence of complex interaction effects between EPs.

Line-Level Aggregation

For each skill $j \in \{1, \ldots, K\}$, the resulting coefficients β_{ij} are used to generate the line-level $EPSPI_n$, which is the arithmetic mean of the four predicted probabilities:

$$\text{EPSPI}_n = \frac{1}{K} \sum_{j=1}^{K} \hat{p}_{nj} \tag{34}$$

This scalar value represents the predicted probability of higher-order skill deployment based on the specific combination of teaching practices observed in line n.

4.4 Method D_GLMM: (Generalised Linear Mixed Model)

Method C fits a fixed-effects logistic regression at the line level, treating all $N = 19{,}181$ lines as independent. This assumption is violated: lines within the same episode share a common classroom context — teacher style, lesson topic, student cohort — that induces within-episode correlation in skill observations Y_{nj}. Ignoring this correlation does not bias the fixed-effect estimates $\hat{\beta}_{ij}$ but does underestimate their standard errors, inflating apparent precision [21, 22]. Method F addresses this by adding an **episode-level random intercept** to the logistic model, explicitly modelling the between-episode variance in baseline skill rates.

Model specification. For each skill S_j, the GLMM is:

$$\text{logit}\, P(Y_{nj} = 1 \mid X_{n\cdot},\, u_{ej}) = \underbrace{\beta_{0j} + \sum_{i=1}^{P} \beta_{ij}\, X_{ni}}_{\text{fixed effects}} + \underbrace{u_{ej}}_{\text{random intercept}}, \tag{35}$$

where $u_{ej} \overset{\text{iid}}{\sim} \mathcal{N}(0,\, \sigma_j^2)$ is the random intercept for episode e and skill j, capturing unobserved episode-level heterogeneity in skill baseline rates. The random intercept u_{ej} is not a free parameter to be estimated per episode, it is a latent variable integrated out during maximum likelihood estimation via Gauss–Hermite quadrature [**?**]. The free parameters are $\{\beta_{0j}, \beta_{1j}, \beta_{2j}, \beta_{3j}, \sigma_j^2\}$ for each skill j, giving $(P + 2) \times K = 20$ free parameters in total ($P + 1$ fixed effects plus 1 variance component per skill).

Relationship to Method C. Method D is a direct extension of Method C. The fixed-effect parameters β_{ij} in Method D have the same interpretation as in Method C — log-odds contribution of EP_i to skill S_j — but they are now estimated *conditional on* the random intercept u_{ej}, making them **subject-specific** (conditional) rather than **population-average** (marginal) effects [24]. In practical terms:

- Method C $\hat{\beta}_{ij}$: the average log-odds increase in $P(Y_{nj} = 1)$ when EP_i is activated, averaged over all episodes.
- Method D $\hat{\beta}_{ij}$: the log-odds increase for a *specific episode* holding its random intercept u_{ej} fixed — a larger effect size than the marginal estimate because it is not attenuated by between-episode variance.

The intraclass correlation coefficient (ICC) quantifies the proportion of total skill variance attributable to between-episode differences:

$$\mathrm{ICC}_j = \frac{\sigma_j^2}{\sigma_j^2 + \pi^2/3}, \tag{36}$$

where $\pi^2/3 \approx 3.29$ is the variance of the standard logistic distribution [**?**]. A high ICC (e.g. ICC > 0.10) indicates that episode-level clustering is substantial and that ignoring it in Method C materially affects inference.

Line-level score. At deployment, the random intercept u_{ej} is an unobserved latent variable. The marginal predicted probability for line n, integrating out u_{ej}, does not have a closed form for the logistic link. Following standard practice [21], we use the **marginal prediction** obtained by setting $u_{ej} = 0$ (the population mean):

$$\mathrm{EPSPI}_n^{(F)} = \frac{1}{K} \sum_{j=1}^{K} \sigma\left(\hat{\beta}_{0j} + \sum_{i=1}^{P} \hat{\beta}_{ij} X_{ni} \right), \tag{37}$$

which is structurally identical to Equation (30) for Method C, with the difference that the $\hat{\beta}_{ij}$ values are now estimated under the GLMM likelihood that accounts for within-episode clustering. Setting $u_{ej} = 0$ gives the prediction for an "average episode" — a new episode drawn from the population with no prior knowledge of its specific classroom context.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(D)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(D)}, \tag{38}$$

where n_e is the actual number of lines in episode e. Because u_{ej} is set to zero in Equation (75), lines within the same episode that share identical EP flags produce identical $\mathrm{EPSPI}_n^{(D)}$ values. Within-episode variation in $\mathrm{EPSPI}_n^{(D)}$ arises only from line-to-line variation in the EP profile $[X_{n1}, X_{n2}, X_{n3}]$.

Why Method D is in-sample only. Method D uses `BinomialBayesMixedGLM` from `statsmodels` [**?**], which fits the full random-effects likelihood via Laplace approximation with Gaussian priors on the fixed effects. Refitting this model 25 times (5 folds ×

5 CV repeats) would require approximately 20 minutes of computation with no expected gain in information: because the marginal prediction sets $u_{ej} = 0$, the CV prediction function for Method F is structurally identical to that of Method E ($u_{ej} = 0$ in both). The CV Pearson r for Method D is therefore expected to equal that of Method C (≈ 0.400) to within Monte Carlo noise, and Method F is reported as *in-sample only* in the results table. Its contribution to the paper is not predictive performance but the ICC values, which quantify the episode-level clustering that motivates the GEE working-correlation specification in Method I.

Quasi-separation and convergence guard. As in Method C, Constructivism's sparsity (1.1% of lines) causes quasi-complete separation in the logistic likelihood [**?**]. A convergence guard $|\hat{\beta}_{ij}| \leq 50$ is applied; coefficients hitting this bound are flagged in the results. Method E (GEE) addresses quasi-separation more principally through the working correlation structure and robust sandwich standard errors, without requiring a hard coefficient bound.

What "Train Pearson r" means for Method D. Method D has $(P + 2) \times K = 20$ free parameters estimated on all N lines. The in-sample Train Pearson r reflects genuine in-sample fit under the GLMM likelihood, with the same caveat as Method E: the risk of overfitting is negligible at the line level ($N = 19{,}181 \gg 20$ parameters) but meaningful at the episode level (49 episodes). Because the marginal prediction $u_{ej} = 0$ is used, the episode-level EPSPI for Method F is expected to be close to that of Method C, and any difference in Train r or Exp.Var between the two methods reflects the change in $\hat{\beta}_{ij}$ induced by accounting for within-episode clustering — not a change in the prediction function's structure.

Properties. Method D has $(P + 2) \times K = 20$ **free parameters** — four more than Method C, corresponding to the $K = 4$ random-effect variance components σ_j^2. It is the only method in the framework that explicitly models within-episode skill clustering. Its primary role is to confirm that Method C's fixed-effect estimates $\hat{\beta}_{ij}$ are not materially distorted by ignoring within-episode correlation; if Method C and Method D produce similar EPSPI_e values and ICC values are low (< 0.05), within-episode clustering is not a confound for the other methods. Method H (NMF) and Method E (GEE) take complementary approaches to the same clustering problem through latent-variable decomposition and working-correlation robustness respectively.

GLMM Estimation Strategies: Laplace Approximation vs. Variational Bayes

Because the Generalised Linear Mixed Model (GLMM) incorporates an unobserved latent variable (the random intercept u_{ej}), the marginal likelihood integral over the logistic link function lacks a closed-form solution. To ensure the robustness of the Method F estimation,

we evaluate two distinct numerical strategies for approximating this integral: the Laplace Approximation (Maximum A Posteriori) and Variational Bayes.

Mathematical Formulations

Laplace Approximation (Maximum A Posteriori): To address the violation of the independence assumption inherent in standard logistic regression (Method C), Method F introduces a Generalised Linear Mixed Model (GLMM). This approach explicitly accounts for within-episode correlation induced by shared classroom contexts (e.g., teacher style, lesson topic) by incorporating an episode-level random intercept.

Mathematical Formulation: The Latent Variable Approach

For a specific line n, in episode e, for skill j, let Y_{nej} be the binary observed outcome (1 if skill present, 0 otherwise). We assume a continuous, unobserved latent variable Y^*_{nej} representing the "propensity" to deploy a skill:

$$Y^*_{nej} = \underbrace{\mathbf{X}_n^T \boldsymbol{\beta}_j}_{\text{Fixed Effects}} + \underbrace{u_{ej}}_{\text{Random Effect}} + \underbrace{\epsilon_{nej}}_{\text{Error}} \tag{39}$$

Where:

- $\mathbf{X}_n^T = [1, X_{n,\text{Beh}}, X_{n,\text{Cog}}, X_{n,\text{Con}}]$ represents the active Educational Practices (EPs).
- $\boldsymbol{\beta}_j = [\beta_{0j}, \beta_{1j}, \beta_{2j}, \beta_{3j}]^T$ are the population-average fixed effects.
- $u_{ej} \sim \mathcal{N}(0, \sigma^2_{uj})$ is the episode-specific random intercept capturing unobserved context.
- $\epsilon_{nej} \sim \text{Logistic}(0, 1)$ is the line-level residual error.

Because the cumulative distribution function of the standard logistic distribution is the sigmoid function, the probability of observing the skill becomes:

$$P(Y_{nej} = 1 \mid \mathbf{X}_n, u_{ej}) = \frac{1}{1 + \exp(-(\mathbf{X}_n^T \boldsymbol{\beta}_j + u_{ej}))} \tag{40}$$

Parameter Estimation via Laplace Approximation

The random effect u_{ej} is an unobserved latent variable. To estimate $\boldsymbol{\beta}_j$ and σ^2_{uj}, we must maximize the marginal likelihood by integrating out u_{ej}:

$$L_{\text{marginal}}(\boldsymbol{\beta}, \sigma^2_u) = \int_{-\infty}^{\infty} \left[\prod_{n \in n_e} P(Y_{nj} \mid \mathbf{X}_n, \boldsymbol{\beta}, u) \right] P(u \mid \sigma^2_u)\, du \tag{41}$$

Due to the logistic link, this integral lacks a closed-form solution. We employ the **Laplace Approximation** (Maximum A Posteriori estimation), which numerically approximates the integral by finding the peak (mode) of the posterior distribution and utilizing the Hessian matrix to estimate the curvature (variance).

Intraclass Correlation Coefficient (ICC) and Marginal Prediction

The Intraclass Correlation Coefficient (ICC) measures the proportion of the total variance in the latent propensity Y^* attributable to the episode-level random effect:

$$\text{ICC}_j = \frac{\text{Var}(u_{ej})}{\text{Var}(Y^*_{nej} \mid \mathbf{X}_n)} = \frac{\sigma^2_{uj}}{\sigma^2_{uj} + \frac{\pi^2}{3}} \tag{42}$$

where $\pi^2/3 \approx 3.29$ is the variance of the standard logistic distribution.

For line-level EPSPI prediction, the framework integrates out the unobserved random effect by setting it to its expected population value, $\mathbb{E}[u_{ej}] = 0$:

$$\text{EPSPI}_n = \frac{1}{K} \sum_{j=1}^{K} \sigma(\mathbf{X}_n^T \hat{\boldsymbol{\beta}}_j) \tag{43}$$

Implications of Episode-Level Clustering

The estimation results reveal profound disparities in how different 21st-century skills manifest in the classroom environment.

The "Teacher-Dependent" Skills (High ICC): Problem Solving (ICC = 0.6552) and Critical Thinking (ICC = 0.5370) exhibit massive episode-level clustering. This indicates that over half of the variance in whether these skills are observed depends on the *unobserved classroom context* (the specific teacher, topic, or cohort) rather than the active learning theory. Even if a teacher switches to Constructivism, their baseline "Episode Personality" ($\sigma_u = 2.5002$ for Problem Solving) heavily dictates the outcome.

The "Pedagogy-Driven" Skills (Low ICC): Conversely, Collaborative Learning (ICC = 0.1237) and Creativity (ICC = 0.1462) show relatively little between-episode variance. Specifically for Collaborative Learning, the fixed-effect coefficient for Constructivism is exceptionally large ($\beta_3 = 7.1466$). The combination of a massive fixed effect and a low ICC indicates a near-universal, structural relationship: when Constructivism is deployed, Collaborative Learning almost universally follows, regardless of the specific teacher or classroom context.

These results mathematically validate the need for hierarchical or robust estimation methods (such as GEE in Method I) when analysing classroom transcripts, as assuming

line-level independence (Method C) critically underestimates the standard errors for teacher-dependent skills.

The Laplace method provides a deterministic, point-estimate approximation of the integral. It locates the mode (peak) of the posterior distribution, denoted as $\hat{\boldsymbol{\theta}} = \{\hat{\boldsymbol{\beta}}, \hat{\sigma}_u^2\}$, and approximates the surrounding area using a second-order Taylor expansion (a Gaussian parabola). The integral is approximated using the determinant of the negative Hessian matrix $\mathbf{H}$ evaluated at the mode:

$$\int P(\mathbf{Y} \mid \boldsymbol{\theta}) P(\boldsymbol{\theta})\, d\boldsymbol{\theta} \approx P(\mathbf{Y} \mid \hat{\boldsymbol{\theta}}) P(\hat{\boldsymbol{\theta}}) \sqrt{\frac{(2\pi)^d}{|-\mathbf{H}|}} \tag{44}$$

Because Laplace relies on the curvature (Hessian) at a single maximum, it is highly precise but computationally brittle when the posterior distribution is highly skewed or flat—such as under conditions of quasi-separation caused by sparse variables (e.g., Constructivism).

Variational Bayes (VB): Instead of finding a single point estimate, Variational Bayes seeks a full distributional approximation. It posits a family of simpler, factorized probability distributions $q(\boldsymbol{\theta})$ and optimizes their parameters to minimize the Kullback-Leibler (KL) divergence to the true, intractable posterior $P(\boldsymbol{\theta} \mid \mathbf{Y})$. This is mathematically equivalent to maximizing the Evidence Lower Bound (ELBO):

$$\text{ELBO}(q) = \mathbb{E}_q[\log P(\mathbf{Y}, \boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})] \leq \log P(\mathbf{Y}) \tag{45}$$

The fixed effects and variance components reported under VB are the means of the optimized approximate distributions $q^*(\boldsymbol{\beta})$ and $q^*(\log \sigma_u^2)$. Because VB integrates over a region rather than evaluating a single point, it applies an inherent regularization effect, making it highly robust to separation issues but potentially smoothing over sharp empirical contrasts.

Empirical Comparison

The two estimation methods were applied to the corpus, yielding diverging estimates for the episode-level standard deviation (σ_u) and the resulting Intraclass Correlation Coefficient (ICC). The comparative results are presented in Table 8.

4.4.1 Variational Bayes VS Laplace Approximation Estimation Results

Table 6 presents the fixed effects and variance components estimated via the Variational Bayes algorithm (Method F). The overall line-level EPSPI distribution yielded a mean $\mu = 0.0753$ and standard deviation $\sigma = 0.0391$. The Laplace Approximation successfully converged for all four skills, yielding the fixed effects, random effect standard deviations

(σ_u), and ICCs detailed in Table 5. The overall line-level EPSPI distribution yielded a mean $\mu = 0.0843$ and standard deviation $\sigma = 0.0472$.

Table 5: GLMM Fixed Effects ($\boldsymbol{\beta}$) and Episode Variance Components via Laplace Approximation

Skill (j)	**Intercept**	**Beh** (β_1)	**Cog** (β_2)	**Con** (β_3)	$\boldsymbol{\sigma_u}$	**ICC**
Problem Solving	−4.0222	1.3236	1.1679	3.0082	2.5002	0.6552
Creativity	−6.8149	−0.3037	0.4950	2.3431	0.7505	0.1462
Collaborative Learn.	−8.9586	3.1086	−0.8171	7.1466	0.6816	0.1237
Critical Thinking	−2.3919	−1.2590	1.8416	1.7063	1.9533	0.5370

Table 6: GLMM Fixed Effects ($\boldsymbol{\beta}$) and Episode Variance Components via Variational Bayes

Skill (j)	**Intercept**	**Beh** (β_1)	**Cog** (β_2)	**Con** (β_3)	$\boldsymbol{\sigma_u}$	**ICC**
Problem Solving	−4.4539	1.3312	1.1758	3.0586	2.2272	0.6013
Creativity	−8.5275	−0.4461	0.3761	1.8237	1.7127	0.4713
Collaborative Learn.	−11.0602	3.2793	−1.0144	6.2312	1.8214	0.5021
Critical Thinking	−2.4040	−1.2649	1.8488	1.7164	1.1497	0.2866

Predictive Performance: Laplace vs. Variational Bayes

Table 7 compares the episode-level predictive performance and line-level information criteria of the two approximation strategies. The Laplace Approximation demonstrates superior out-of-sample generalisation, achieving a higher Cross-Validated Pearson r and lower Cross-Validated Mean Squared Error. In fact, it performs better at every metric except at the line level information criteria (AIC/BIC), where Variational Bayes performs better.

Table 7: Predictive Performance Comparison of GLMM Approximations

Method	**EPSPI Mean**	**Train** r	**Train** R^2	**Exp.Var (cal)**	**Exp.Var (raw)**	**CV** r	**CV MSE**	**AIC (line)**	**BIC (line)**
D_GLMM (Laplace)	0.0920	0.6714	0.4508	0.4508	0.4033	0.4354	0.0134	46501.800000	46604.000000
D_GLMM (VB)	0.0799	0.6425	0.4128	0.4128	0.2957	0.4005	0.0162	46134.1	46236.3

4.4.2 Interpretations and Methodological Implications

The divergence between the Laplace and VB estimates provides deep insight into the structure of the pedagogical data, specifically regarding how statistical methods handle sparse, highly predictive covariates.

Table 8: Comparison of GLMM Variance Components: Laplace vs. Variational Bayes

	Laplace Approximation		**Variational Bayes**	
Skill (j)	$\boldsymbol{\sigma_u}$	**ICC**	$\boldsymbol{\sigma_u}$	**ICC**
Problem Solving	2.5002	0.6552	2.2272	0.6013
Critical Thinking	1.9533	0.5370	1.1497	0.2866
Collaborative Learn.	0.6816	0.1237	1.8214	0.5021
Creativity	0.7505	0.1462	1.7127	0.4713
Line-level Aggregation	$\mu = 0.0843$	$\sigma = 0.0472$	$\mu = 0.0753$	$\sigma = 0.0391$

1. The "Variance Trade-off" in Quasi-Separated Data: Under the Laplace approximation, Collaborative Learning and Creativity exhibited very low ICCs (≈ 0.12–0.14). As noted previously, this occurs because the model attributes nearly all the variance to the massive fixed effect of Constructivism ($\beta_{\text{Con}} \approx 7.14$). Conversely, Variational Bayes shrinks these extreme fixed-effect coefficients toward the prior mean to maintain a stable probability distribution. Because the fixed effect is artificially restricted by the VB prior, the model is forced to attribute the unexplained pedagogical variance to the random intercept u_e, artificially inflating the ICC for these skills to ≈ 0.47–0.50.

2. Polarization vs. Smoothing: The Laplace method provides a polarized view of the classroom: skills are either highly teacher-dependent (Problem Solving, ICC = 0.65) or almost entirely pedagogy-driven (Collaborative Learning, ICC = 0.12). Variational Bayes smooths this landscape, pulling all ICCs toward the center (0.28–0.60). Furthermore, VB results in a more compressed line-level EPSPI distribution ($\sigma = 0.0391$ vs Laplace $\sigma = 0.0472$), reflecting its inherent Bayesian regularization.

3. Conclusion for Method D: While Variational Bayes provides superior algorithmic stability against the quasi-complete separation caused by Constructivism, the Laplace Approximation provides a more theoretically accurate reflection of the maximum likelihood estimates. The Laplace results confirm that while some skills are heavily clustered by episode, structural pedagogical links (such as Constructivism driving Collaborative Learning) remain robust independently of classroom context.

4.5 Method E: Generalised Estimating Equations (GEE)

Method E estimates population-average EP–skill associations by treating all N lines as independent. Method D corrects for within-episode clustering through a subject-specific random intercept, but the marginal prediction still sets $u_{ej} = 0$, making its deployment prediction identical in structure to Method C. Method E takes a third approach: it directly models the **within-episode correlation structure** of the binary skill observations Y_{nj}

through a working correlation matrix, producing population-average coefficient estimates that are consistent even if the working correlation is mis-specified [23, 24]. GEE does not require a full likelihood — it solves a set of generalised estimating equations that generalise the score equations of logistic regression to the correlated data setting [**?**].

Model specification. For each skill S_j, GEE specifies the same marginal mean model as Method C:

$$\operatorname{logit} \mu_{nj} = \operatorname{logit} E[Y_{nj} \mid X_{n\cdot}] = \beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni}, \tag{46}$$

with the additional specification of a **working correlation matrix** $\mathbf{R}(\alpha_j)$ for the n_e binary outcomes within episode e:

$$\operatorname{Corr}(Y_{nj}, Y_{n'j} \mid e_n = e_{n'} = e) = \alpha_j, \quad n \neq n', \tag{47}$$

the exchangeable structure, which assumes all pairs of lines within the same episode have the same correlation α_j. The exchangeable structure is chosen because there is no natural ordering of lines within an episode that would justify an autoregressive or unstructured matrix. GEE estimates $\hat{\beta}_{ij}$ and $\hat{\alpha}_j$ iteratively via the **iteratively reweighted least squares** (IRLS) algorithm, alternating between updating $\hat{\beta}_{ij}$ with current $\hat{\alpha}_j$ fixed and updating $\hat{\alpha}_j$ from the Pearson residuals [23]. The final standard errors use the **sandwich (robust) variance estimator** $V_{\text{sand}} = A^{-1}BA^{-1}$, which is consistent regardless of whether the working correlation $\mathbf{R}(\hat{\alpha}_j)$ is correctly specified [**?**].

Relationship to Methods C and D All three methods — C, D, and I — estimate the same population-average log-odds:

$$\beta_{ij} = \operatorname{logit} P(Y_{nj} = 1 \mid X_{ni} = 1,\, X_{ni'} = 0 \,\forall\, i' \neq i) - \beta_{0j}.$$

They differ in how within-episode correlation is handled during estimation:

- **Method C**: assumes $\alpha_j = 0$ (independence). Consistent if $\alpha_j = 0$; standard errors too small if $\alpha_j > 0$.
- **Method D**: models α_j through a random intercept variance σ_j^2; gives conditional (subject-specific) not marginal estimates.
- **Method E**: estimates α_j directly as a nuisance parameter; gives marginal estimates with sandwich standard errors valid for any α_j.

The estimated within-episode correlation $\hat{\alpha}_j$ is itself a substantive finding: it quantifies how much two randomly chosen lines from the same episode resemble each other in skill expression, above and beyond what is predicted by their EP flags.

Three-stage convergence fallback. Constructivism appears in only 1.1% of lines, causing quasi-complete separation in the GEE score equations — the same problem as Methods C and D. Additionally, the exchangeable working correlation occasionally fails to converge for rare EP profiles. A three-stage fallback is implemented:

1. **Exchangeable** working correlation with $|\hat{\beta}_{ij}| \leq 50$ guard.
2. If exchangeable diverges: **independence** working correlation (reduces to Method C estimates with sandwich standard errors).
3. If independence also diverges: **global θ_{ij} fallback** — the Laplace-smoothed frequency estimate from Equation (**??**).

The fallback stage used for each skill j is recorded and reported in Appendix **??**. For the exchangeable structure, the estimated correlation coefficients across the four skills are: $\hat{\alpha}_{\text{PS}} = 0.450$, $\hat{\alpha}_{\text{Cr}} = 0.043$, $\hat{\alpha}_{\text{CL}} = 0.223$, $\hat{\alpha}_{\text{CT}} = 0.021$, indicating moderate within-episode clustering for Problem Solving and weak clustering for the remaining skills.

Line-level score. The Method E score for line n is the mean marginal predicted probability across all K skills:

$$\text{EPSPI}_n^{(E)} = \frac{1}{K} \sum_{j=1}^{K} \sigma\left(\hat{\beta}_{0j}^{(E)} + \sum_{i=1}^{P} \hat{\beta}_{ij}^{(E)} X_{ni} \right), \tag{48}$$

where $\hat{\beta}_{ij}^{(E)}$ are the GEE-estimated coefficients and $\sigma(\cdot)$ is the logistic function. The formula is structurally identical to Methods C and D (Equations (30) and (75)) — the difference from C and D lies entirely in the coefficient values $\hat{\beta}_{ij}^{(E)}$, which are estimated under the exchangeable working correlation rather than under independence or a random-effects likelihood.

Example: line with only Constructivism active. Setting $X_{n1} = 0$, $X_{n2} = 0$, $X_{n3} = 1$:

$$\text{EPSPI}_n^{(E)} = \frac{1}{4} \sum_{j=1}^{4} \sigma\left(\hat{\beta}_{0j}^{(E)} + \hat{\beta}_{3j}^{(E)} \right).$$

The GEE coefficients $\hat{\beta}_{3j}^{(E)}$ will differ from the logistic $\hat{\beta}_{3j}^{(C)}$ because GEE downweights large episodes (high n_e) relative to small episodes when the working correlation is exchangeable — a line in a 587-line episode contributes less to the estimating equations than a line in a 52-line episode, adjusting for the fact that lines in large episodes are more correlated with each other.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(E)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n=e\}} \mathrm{EPSPI}_n^{(E)}, \tag{49}$$

where n_e is the actual number of lines in episode e. As with Methods C and D, $\mathrm{EPSPI}_n^{(E)}$ varies line by line through the EP profile $[X_{n1}, X_{n2}, X_{n3}]$, so the episode mean is a genuine average of heterogeneous line scores rather than a constant.

What "Train Pearson r" means for Method I. Method I has $K \times (P+1) = 16$ free parameters $\hat{\beta}_{ij}^{(E)}$ plus $K = 4$ nuisance correlation parameters $\hat{\alpha}_j$, giving 20 estimated quantities in total. The $\hat{\alpha}_j$ are nuisance parameters — they improve the efficiency of $\hat{\beta}_{ij}^{(E)}$ but are not used in the prediction formula (48). The in-sample Train Pearson r reflects genuine in-sample fit with the same episode-level overfitting caveat as Method E: $\hat{\beta}_{ij}^{(E)}$ is estimated from all 49 episodes' line-level Y_{nj}, so the correlation between $\mathrm{EPSPI}_e^{(E)}$ and $\bar{Y}_e$ is not a held-out evaluation. Method E achieves the highest CV $r = 0.450$ and the lowest CV MSE $= 0.008018$ of all eleven methods, confirming that accounting for within-episode correlation in estimation genuinely improves out-of-sample predictive validity — not merely in-sample fit.

Properties. Method E has $K \times (P+1) = 16$ **free parameters** in the prediction formula, identical in count to Method C, with an additional $K = 4$ nuisance correlation estimates. It achieves the best CV performance of all eleven methods and is therefore the **recommended method** for predicting episode-level skill deployment from EP structure. The key advantages over Method E are: (1) standard errors are valid under any within-episode correlation structure; (2) coefficient estimates are more efficient when $\hat{\alpha}_j > 0$; (3) the sandwich variance estimator provides a robustness check on the logistic functional form assumption. The key advantage over Method D is full cross-validatability: unlike the GLMM likelihood, GEE estimating equations can be solved on 39 training episodes in under one second per fold, making 5-fold CV computationally trivial. Its limitation relative to Method D is that GEE does not provide episode-specific random effect estimates $\hat{u}_{ej}$ and therefore cannot answer questions about individual episode-level deviations from the population average — the non-EP residual $\varepsilon_e = \bar{Y}_e - \mathrm{EPSPI}_e^{(E)}$ serves this diagnostic role instead (Section **??**).

4.6 Method F: Principal Component Analysis

Methods A, B, C, D, and E all work directly with the $P \times K$ conditional probability matrix Θ, using either raw frequencies or parametric log-odds models. Method F takes a different approach: rather than modelling θ_{ij} directly, it asks whether the 12-dimensional EP–skill

Table 9: GEE estimated coefficients $\hat{\beta}^{GEE}$ with robust 95% Wald confidence intervals (logit scale). Asterisks (*) indicate confidence intervals that exclude zero (significant at the 5% level).

Skill	Method	Intercept	$\hat{\beta}_{Beh}$	$\hat{\beta}_{Cog}$	$\hat{\beta}_{Con}$
Problem Solving	Exchangeable	-1.678 [-2.484, -0.873]*	0.655 [0.027, 1.283]*	0.568 [0.033, 1.103]*	1.851 [0.381, 3.322]*
Creativity	Exchangeable	-5.463 [-7.037, -3.890]*	-0.100 [-1.792, 1.592]	0.694 [-0.696, 2.085]	2.076 [0.716, 3.435]*
Collab. Learning	Independence	-5.362 [-7.605, -3.119]*	1.462 [0.629, 2.295]*	-0.516 [-1.073, 0.041]	6.907 [5.136, 8.678]*
Critical Thinking	Exchangeable	-2.129 [-2.928, -1.330]*	-0.984 [-1.256, -0.712]*	1.558 [1.065, 2.051]*	1.436 [0.721, 2.150]*

profile space contains a lower-dimensional **latent structure** that summarises the essential variation in EP–skill co-occurrence patterns. Principal Component Analysis (PCA) finds this structure by identifying the directions of maximum variance in the column space of Θ, without imposing any constraint on the sign or magnitude of the components [26**?**].

Input matrix and centring. PCA is applied to the global Θ matrix. The 3×4 matrix $\Theta = [\theta_{ij}]$ has rows indexed by EP ($i = 1, 2, 3$) and columns indexed by skill ($j = 1, 2, 3, 4$). Before decomposition, Θ is column-centred:

$$\tilde{\theta}_{ij} = \theta_{ij} - \bar{\theta}_{\cdot j}, \qquad \bar{\theta}_{\cdot j} = \frac{1}{P} \sum_{i=1}^{P} \theta_{ij}, \tag{50}$$

so that each skill column has zero mean across the three EPs. Centring ensures PCA captures *differential* EP–skill affinity — how each EP deviates from the average skill profile — rather than the overall level of skill occurrence.

Decomposition. The singular value decomposition (SVD) of the centred matrix $\tilde{\Theta} \in \mathbb{R}^{P \times K}$ is:

$$\tilde{\Theta} = \mathbf{U}\,\mathbf{S}\,\mathbf{V}^{\top}, \tag{51}$$

where $\mathbf{U} \in \mathbb{R}^{P \times P}$ contains the left singular vectors (EP loadings), $\mathbf{S}$ is diagonal with singular values $s_1 \geq s_2 \geq s_3 \geq 0$, and $\mathbf{V} \in \mathbb{R}^{K \times P}$ contains the right singular vectors (skill loadings). With $P = 3$ EPs and $K = 4$ skills, at most $\min(P, K) = 3$ components are available. Method F retains $r = 1$ component on parsimony grounds: the NMF rank analysis (Section 4.8) confirmed that $r = 2$ adds negligible signal ($|r(\delta_{\text{Beh}}, \bar{Y}_e)| = 0.013$, $|r(\delta_{\text{Cog}}, \bar{Y}_e)| = 0.144$).

EP scores on the first component. The score of each EP on the first principal component is:

$$\phi_i^{(1)} = s_1\, u_{i1}, \quad i = 1, 2, 3, \tag{52}$$

where u_{i1} is the i-th element of the first left singular vector. $\phi_i^{(1)}$ measures how strongly EP_i is associated with the dominant variance direction. Because Constructivism has

substantially higher conditional probabilities than Behaviourism and Cognitivism ($\theta_{3j} \gg \theta_{1j} \approx \theta_{2j}$ for all j), the first component is dominated by the Constructivism–other contrast: $\phi^{(1)}_{\text{Con}} \gg \phi^{(1)}_{\text{Beh}} \approx \phi^{(1)}_{\text{Cog}}$.

Line-level score. The Method F score for line n is the projection of the line's EP activation vector onto the first principal component, normalised by p_n:

$$\text{EPSPI}^{(F)}_n = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni}\, \phi^{(1)}_i, \qquad p_n > 0, \tag{53}$$

where $p_n = \sum_i X_{ni}$ is the number of active EPs. For a line with only Constructivism active ($p_n = 1$, $X_{n3} = 1$): $\text{EPSPI}^{(F)}_n = \phi^{(1)}_{\text{Con}}$ — a large positive value. For Behaviourism-only or Cognitivism-only lines, $\text{EPSPI}^{(F)}_n \approx 0$ after centring. When $p_n = 0$, the fallback $\text{EPSPI}^{(F)}_n = 0$ (the centred mean) is used.

Episode-level aggregation.

$$\text{EPSPI}^{(F)}_e = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \text{EPSPI}^{(F)}_n, \tag{54}$$

where n_e is the actual number of lines in episode e. Because 47 of 49 episodes have $\delta^e_{\text{Con}} \approx 0$, $\text{EPSPI}^{(F)}_e$ clusters near zero for almost all episodes, with large positive values only for Episodes 13 and 49 — the high-leverage Constructivist episodes identified in Section 5.2. This compression is the direct cause of the negative Exp.Var(raw) for Method C: raw predictions cluster near zero while $\bar{Y}_e \approx 0.20$, producing systematic positive residuals $\bar{Y}_e - \text{EPSPI}^{(F)}_e > 0$ for almost every episode.

Why Exp.Var(cal) is positive despite negative Exp.Var(raw). The OLS calibration $\hat{Y}_e = a + b \cdot \text{EPSPI}^{(F)}_e$ with fitted slope $b \approx 3.2$ expands the compressed PCA scores back to the range of $\bar{Y}_e$, and intercept $a > 0$ shifts them upward. After calibration the *rank ordering* of episodes is preserved — Method F correctly identifies that high-Constructivism episodes have higher skill rates — and Exp.Var(cal) ≈ 0.49 reflects this preserved ordering. The gap Exp.Var(cal) − Exp.Var(raw) directly quantifies the scale bias from PCA compression: a method with zero scale bias satisfies Exp.Var(cal) = Exp.Var(raw).

Relationship to θ_{ij} and Methods A, B Method F is a **rank-1 approximation** to Θ:

$$\hat{\Theta}^{(1)} = s_1\, \mathbf{u}_1\, \mathbf{v}_1^{\top}, \tag{55}$$

replacing the full 3×4 Θ matrix with its best rank-1 reconstruction. Method A uses the full Θ directly; Method F discards the second and third components. The CV r gap between

Methods A and F (0.409 vs 0.281) quantifies how much multi-dimensional structure in Θ is lost by the rank-1 constraint — specifically, the Behaviourism vs Cognitivism contrast (second component) carries additional predictive signal that Method A uses but Method F discards.

What "Train Pearson r" means for Method F Method F has **1 free parameter** — the first principal component direction $\mathbf{u}_1$, estimated from Θ which was itself computed from all N lines. In cross-validation, Θ is recomputed from 39 training episodes, the SVD is re-run to obtain new $\phi_i^{(1)}$ values, and $\text{EPSPI}_e^{(F)}$ is computed for the 10 test episodes using these training-derived component scores. The larger Train-to-CV drop for Method F compared to Methods A and B reflects additional estimation variance from the SVD decomposition of a 3×4 matrix estimated from only 39 episodes.

Properties. Method F has **1 free parameter** and is the first method on the Dimensionality Reduction axis. Its key advantage is robustness to correlated EP–skill columns: if two skills S_j and $S_{j'}$ are highly correlated across EPs, PCA consolidates them into a single component rather than treating them as independent signals. Its key limitation is scale compression caused by the Constructivism dominance of the first component — Behaviourism and Cognitivism lines receive near-zero scores regardless of their EP flags, making raw predictions uninformative for the 98.9% of lines with no Constructivism activation. Method H (NMF) addresses this by imposing non-negativity, preventing Behaviourism and Cognitivism from being suppressed in the component scores, while Method G (Correspondence Analysis) works in the chi-squared metric of the EP–skill contingency table rather than the Euclidean metric of Θ, giving a more balanced weighting to rare EPs.

4.7 Method G: Correspondence Analysis

Method F applies PCA to the conditional probability matrix Θ in the Euclidean metric, treating each cell θ_{ij} as equally important. This gives disproportionate influence to Constructivism — whose θ_{3j} values are far from the row mean — while suppressing the subtler Behaviourism–Cognitivism contrast. Method G addresses this through **Correspondence Analysis** (CA), which decomposes the EP–skill *contingency table* in the **chi-squared metric**, weighting each cell by its deviation from statistical independence relative to its expected frequency [27**?**]. CA is the natural multivariate method for binary co-occurrence data: it simultaneously finds the low-dimensional representation of EPs and skills that best recovers the association structure of the contingency table, giving proportionally more weight to rare EP–skill combinations than PCA does.

Input matrix: the contingency table. CA operates on the raw co-occurrence count matrix $\mathbf{N} \in \mathbb{Z}^{P\times K}$, where:

$$N_{ij} = \sum_{n=1}^{N} X_{ni}\, Y_{nj}, \tag{56}$$

is the number of lines in which EP_i and S_j co-occur simultaneously — the unnormalised numerator of $P(\mathrm{EP}_i \cap S_j)$ from Equation (**??**). Unlike PCA, CA does not require column-centring because the chi-squared metric already centres the decomposition around the independence model.

Chi-squared decomposition. Let $r_i = N_{i\cdot}/N_{\cdot\cdot}$ be the row mass (marginal EP proportion) and $c_j = N_{\cdot j}/N_{\cdot\cdot}$ be the column mass (marginal skill proportion), where $N_{\cdot\cdot} = \sum_{i,j} N_{ij}$. Note that $r_i = P(\mathrm{EP}_i)$ and $c_j = P(S_j)$ from Equation (**??**). CA decomposes the matrix of standardised residuals from independence:

$$s_{ij} = \frac{p_{ij} - r_i c_j}{\sqrt{r_i c_j}}, \qquad p_{ij} = \frac{N_{ij}}{N_{\cdot\cdot}} = P(\mathrm{EP}_i \cap S_j), \tag{57}$$

via SVD: $\mathbf{S} = \mathbf{U}\,\mathbf{D}\,\mathbf{V}^\top$, where s_{ij} is the (i,j) element of $\mathbf{S}$ and $d_1 \geq d_2 \geq d_3$ are the singular values. The total inertia $\sum_k d_k^2$ equals the normalised chi-squared statistic $\chi^2/N_{\cdot\cdot}$, measuring total EP–skill association [27].

EP coordinates on the first axis. The standard row coordinates of each EP on the first CA axis are:

$$\psi_i^{(1)} = \frac{u_{i1}\, d_1}{\sqrt{r_i}}, \quad i = 1, 2, 3, \tag{58}$$

where u_{i1} is the i-th element of the first left singular vector. The $1/\sqrt{r_i}$ scaling gives proportionally larger coordinates to rare EPs: Constructivism ($r_{\mathrm{Con}} = 0.011$) receives a much larger coordinate than in PCA, and Behaviourism and Cognitivism are meaningfully separated because their chi-squared residuals s_{ij} differ non-trivially even though their absolute θ_{ij} values are similar.

Sign normalisation. CA coordinates have an arbitrary sign: multiplying all row and column coordinates by -1 produces an equivalent solution. The sign is anchored so that $\psi_{\mathrm{Con}}^{(1)} > 0$, placing Constructivism on the positive side of the first axis consistent with its high skill-conducive profile.

Rescaling to $[0, 1]$. After sign normalisation, $\psi_i^{(1)}$ may be negative for Behaviourism or Cognitivism, producing negative EPSPI scores. To preserve the interpretation of EPSPI

as a skill potential in $[0, 1]$, coordinates are linearly rescaled:

$$\tilde{\psi}_i^{(1)} = \frac{\psi_i^{(1)} - \min_{i'} \psi_{i'}^{(1)}}{\max_{i'} \psi_{i'}^{(1)} - \min_{i'} \psi_{i'}^{(1)}}, \tag{59}$$

which preserves the rank ordering of EPs on the first axis.

Line-level score. The Method G score for line n is:

$$\mathrm{EPSPI}_n^{(G)} = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni}\, \tilde{\psi}_i^{(1)}, \qquad p_n > 0, \tag{60}$$

where $p_n = \sum_i X_{ni}$. For a line with only Constructivism active: $\mathrm{EPSPI}_n^{(G)} = \tilde{\psi}_{\mathrm{Con}}^{(1)} = 1.0$ (after normalisation). For a line with only Behaviourism active: $\mathrm{EPSPI}_n^{(G)} = \tilde{\psi}_{\mathrm{Beh}}^{(1)}$, which under the chi-squared metric reflects the *relative* deviation of Behaviourism's skill profile from independence — more informative than PCA's Euclidean distance from the row mean. When $p_n = 0$, the fallback $\mathrm{EPSPI}_n^{(G)} = 0.5$ (midpoint of $[0, 1]$) is used.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(G)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(G)}, \tag{61}$$

where n_e is the actual number of lines in episode e. Unlike Method F, where almost all episodes received near-zero scores, Method G's $1/\sqrt{r_i}$ scaling amplifies the Behaviourism–Cognitivism contrast, giving meaningfully different $\mathrm{EPSPI}_e^{(G)}$ values across episodes even when Constructivism is absent. This avoids the scale compression that caused Method F's negative Exp.Var(raw).

Relationship to Methods A and F. CA and PCA both decompose a matrix derived from Θ but in different metrics:

- **PCA (Method F)** decomposes $\tilde{\Theta}$ in the Euclidean metric — all cells θ_{ij} equally weighted. Constructivism dominates because its θ_{3j} values are far from the column mean in absolute terms.
- **CA (Method G)** decomposes $\mathbf{S}$ (standardised residuals from independence) in the chi-squared metric — rare EP–skill combinations that deviate from independence are upweighted. Constructivism still has large coordinates, but Behaviourism and Cognitivism are also meaningfully separated through their relative deviations s_{ij}.

CA theory predicts that it finds the representation maximising the correlation between EP row scores and skill column scores simultaneously [27] — exactly the structure that

the logistic AIC rewards. This explains why Method G achieves the **best AIC** (44,358) of all eleven methods: CA directly optimises chi-squared EP–skill co-occurrence at the line level, which is precisely what the logistic likelihood rewards.

The AIC–CV tension. Method G achieves the best AIC but only CV $r = 0.296$ — among the weakest episode-level predictive validity. This tension is a genuine scientific finding, not an artefact. AIC measures **line-level discriminative power**: how well $\text{EPSPI}_n^{(G)}$ separates lines with high vs low skill rates. CV r measures **episode-level predictive validity**: how well $\text{EPSPI}_e^{(G)}$ predicts the aggregate skill rate of unseen episodes. CA optimises the former by construction. But episode-level aggregation (Layer 2) averages out much of this line-level discrimination: an episode with a balanced EP mix will have a moderate $\text{EPSPI}_e^{(G)}$ that loses the within-episode variation that drives the AIC advantage. This AIC–CV tension demonstrates that line-level discriminative power and episode-level predictive validity are genuinely distinct properties — a method can excel at one while underperforming at the other.

What "Train Pearson r" means for Method G. Method G has $P = 3$ **free parameters** — the three EP row coordinates $\tilde{\psi}_i^{(1)}$, estimated from the contingency table $\mathbf{N}$ computed from all N lines. In cross-validation, $\mathbf{N}$ is recomputed from 39 training episodes, the CA decomposition is re-run with sign normalisation and $[0, 1]$ rescaling re-applied, and $\text{EPSPI}_e^{(G)}$ is computed for the 10 test episodes using these training-derived coordinates. The in-sample Train Pearson r is inflated relative to CV r because the CA coordinates are optimised on the same 49 episodes used for evaluation.

Properties. Method G has $P = 3$ **free parameters**. It is the only method whose estimation criterion (chi-squared association) directly corresponds to the logistic AIC used for evaluation, explaining its AIC dominance. Its limitation for episode-level prediction is that the chi-squared metric upweights rare EP–skill combinations that are informative at the line level but average out at the episode level. The sign ambiguity (Equation (59)) and the need for $[0, 1]$ rescaling are minor implementation costs — the substantive CA output is the rank ordering of EPs on the first axis, which is preserved by Equation (59). Together with Methods F and H, Method G completes a systematic comparison of the rank-1 constraint on Θ under three geometric frameworks: Euclidean (PCA), chi-squared (CA), and non-negative (NMF) — examining whether the choice of metric materially affects episode-level predictive validity.

4.8 Method H: Non-Negative Matrix Factorisation (NMF)

Methods F and G apply unconstrained (PCA) and chi-squared (CA) decompositions to Θ, both of which allow negative component values. Negative values are mathematically valid

but pedagogically uninterpretable: a negative loading on Constructivism would imply that Constructivist teaching *reduces* the skill-potential score, which has no meaningful interpretation in the context of educational practice. Method H imposes the **non-negativity constraint** on both the EP factor scores and the skill factor scores, ensuring that the decomposition represents additive combinations of teaching-mode profiles — each component contributes positively to the overall EPSPI score [28, 35].

Input matrix and constraint. NMF is applied directly to the global Θ matrix (not column-centred, unlike PCA):

$$\Theta \approx \mathbf{W}\,\mathbf{H}, \qquad \mathbf{W} \geq 0, \quad \mathbf{H} \geq 0, \tag{62}$$

where $\mathbf{W} \in \mathbb{R}_{\geq 0}^{P \times r}$ contains the EP factor scores (how strongly each EP loads on each latent teaching mode) and $\mathbf{H} \in \mathbb{R}_{\geq 0}^{r \times K}$ contains the skill factor scores (how strongly each latent teaching mode is associated with each skill). The rank r controls the number of latent teaching modes extracted. Both $\mathbf{W}$ and $\mathbf{H}$ are non-negative by constraint, so every element of the approximation $\hat{\Theta} = \mathbf{WH}$ is a *non-negative additive combination* of the r latent profiles. No column-centring is applied because non-negativity and centring are incompatible: centring introduces negative values that the non-negativity constraint would then suppress, distorting the factorisation.

Estimation. For $r = 1$, $\mathbf{W} \in \mathbb{R}_{\geq 0}^{3 \times 1}$ and $\mathbf{H} \in \mathbb{R}_{\geq 0}^{1 \times 4}$, so the factorisation reduces to a rank-1 outer product:

$$\Theta \approx \mathbf{w}\,\mathbf{h}^\top, \qquad \mathbf{w} = [w_1, w_2, w_3]^\top \geq 0, \quad \mathbf{h} = [h_1, h_2, h_3, h_4]^\top \geq 0, \tag{63}$$

where $w_i \geq 0$ is the loading of EP_i on the single latent teaching mode and $h_j \geq 0$ is the loading of skill S_j on that mode. Parameters are estimated by minimising the Frobenius-norm reconstruction error:

$$\min_{\mathbf{w} \geq 0,\, \mathbf{h} \geq 0} \left\| \Theta - \mathbf{w}\,\mathbf{h}^\top \right\|_F^2 = \min_{\mathbf{w},\, \mathbf{h}} \sum_{i=1}^{P} \sum_{j=1}^{K} \left(\theta_{ij} - w_i h_j \right)^2, \tag{64}$$

solved by the multiplicative update rules of [35], which guarantee $\mathbf{w} \geq 0$ and $\mathbf{h} \geq 0$ are maintained throughout optimisation. The observed fitted values for the corpus are:

$$\mathbf{w} = [0.014,\ 0.000,\ 1.000]^\top \quad (\text{Beh},\ \text{Cog},\ \text{Con}),$$

$$\mathbf{h} = [0.779,\ 0.038,\ 0.748,\ 0.346] \quad (\text{PS},\ \text{Cr},\ \text{CL},\ \text{CT}),$$

with reconstruction MSE $= 0.394$. The near-zero $w_{\text{Cog}} = 0.000$ confirms that the single latent mode is almost entirely Constructivism: NMF has identified that Constructivism is the only EP with a substantially different skill profile from the global baseline, and Behaviourism and Cognitivism are effectively indistinguishable at rank 1.

Interpretation of the latent teaching mode. The rank-1 factorisation $\hat{\theta}_{ij} = w_i h_j$ decomposes the EP–skill affinity into two independent factors:

- w_i: how strongly EP_i activates the latent teaching mode. Constructivism ($w_3 = 1.000$) fully activates it; Behaviourism ($w_1 = 0.014$) barely activates it; Cognitivism ($w_2 = 0.000$) does not activate it at all.
- h_j: how strongly the latent mode is associated with skill S_j. Problem Solving ($h_1 = 0.779$) and Collaborative Learning ($h_3 = 0.748$) are most strongly associated; Creativity ($h_2 = 0.038$) is weakest.

The latent teaching mode can therefore be interpreted as a **Constructivist–collaborative skill mode**: it is almost exclusively activated by Constructivism, and it primarily predicts Problem Solving and Collaborative Learning. This aligns with established educational theory: Constructivist pedagogy emphasises active knowledge construction through collaborative problem-solving [36–39].

Line-level score. The Method H score for line n is the NMF-reconstructed conditional skill probability, normalised by p_n:

$$\text{EPSPI}_n^{(H)} = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni} \left(\frac{1}{K} \sum_{j=1}^{K} w_i h_j \right) = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni}\, w_i \cdot \bar{h}, \quad p_n > 0, \tag{65}$$

where $\bar{h} = \frac{1}{K} \sum_{j=1}^{K} h_j$ is the mean skill loading of the latent mode. The inner average $\frac{1}{K} \sum_j w_i h_j = w_i \bar{h}$ factorises into the EP loading w_i and the constant $\bar{h}$, so Method H is equivalent to Method A with θ_{ij} replaced by the rank-1 approximation $\hat{\theta}_{ij} = w_i h_j$.

Example: line with only Constructivism active. Setting $X_{n3} = 1$, $p_n = 1$:

$$\text{EPSPI}_n^{(H)} = w_3 \cdot \bar{h} = 1.000 \times \tfrac{1}{4}(0.779 + 0.038 + 0.748 + 0.346) = 1.000 \times 0.478 = 0.478.$$

Compare to Method A: $\text{EPSPI}_n^{(A)} = 0.510$. The difference (0.032) is the reconstruction error for the Constructivism row — NMF slightly underestimates the Constructivism affinity because it must find a single $\mathbf{w}\mathbf{h}^\top$ that approximates all three EP rows simultaneously under non-negativity.

Example: line with only Cognitivism active. Setting $X_{n2} = 1$, $p_n = 1$:

$$\mathrm{EPSPI}_n^{(H)} = w_2 \cdot \bar{h} = 0.000 \times 0.478 = 0.000.$$

A pure-Cognitivism line receives EPSPI = 0, indicating that Cognitivism contributes nothing to the latent Constructivist–collaborative mode. Compare to Method A: $\mathrm{EPSPI}_n^{(A)} = 0.147$. This is the scale compression that causes $\mathrm{EPSPI}_e^{(H)} \approx 0.028$ to be far below $\bar{Y}_e \approx 0.20$ for most episodes, and is the direct cause of the negative Exp.Var(raw) for Method H. When $p_n = 0$, the fallback $\mathrm{EPSPI}_n^{(H)} = \bar{h}/P = 0.478/3 = 0.159$ is used.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(H)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(H)}, \tag{66}$$

where n_e is the actual number of lines in episode e. Because $w_{\mathrm{Cog}} = 0.000$ and $w_{\mathrm{Beh}} = 0.014 \approx 0$, episodes with no Constructivism lines receive $\mathrm{EPSPI}_e^{(H)} \approx 0$ regardless of their Behaviourism or Cognitivism content. This is the same scale compression as Method F, but for a different reason: in PCA, compression arises from Constructivism dominating the principal component direction; in NMF, it arises from Behaviourism and Cognitivism having near-zero w_i values because the non-negativity constraint prevents them from having large positive loadings on a Constructivism-dominated mode.

Relationship to Methods A, F, and G. All three dimensionality reduction methods (F, G, H) are rank-1 approximations to Θ, differing only in metric and constraint:

Method	Input	Metric	Constraint
F (PCA)	$\tilde{\Theta}$ (centred)	Euclidean	None
G (CA)	$\mathbf{N}$ (counts)	Chi-squared	None
H (NMF)	Θ (uncentred)	Euclidean	$\mathbf{W}, \mathbf{H} \geq 0$

All three produce similar CV r values (0.281–0.348), confirming that the rank-1 constraint is the binding limitation, not the choice of metric or non-negativity. The additional interpretability of NMF — additive teaching-mode profiles with non-negative loadings — comes at no cost in predictive performance relative to PCA or CA. Method A (full Θ, no rank constraint) outperforms all three (CV $r = 0.409$), quantifying the predictive information contained in the rank-2 and rank-3 components of Θ that all dimensionality reduction methods discard.

What "Train Pearson r" means for Method H. Method H has $P + K = 7$ **free parameters** (w_1, w_2, w_3 and h_1, h_2, h_3, h_4), estimated by minimising the Frobenius-norm

reconstruction error on the global Θ matrix. Because Θ is computed from all N lines, the in-sample Train Pearson r reflects in-sample association. In cross-validation, Θ is recomputed from 39 training episodes, NMF is re-run to obtain new $\mathbf{w}$ and $\mathbf{h}$ vectors, and $\text{EPSPI}_e^{(H)}$ is computed for the 10 test episodes using these training-derived factors. The drop from Train $r \approx 0.70$ to CV $r \approx 0.35$ is moderate — comparable to Methods A and B — reflecting the low estimation variance of a $P + K = 7$ parameter model fitted to a $3 \times 4 = 12$-cell matrix.

Properties. Method H has $P + K = 7$ **free parameters** — more than Methods F and G (1 and 3 respectively) but far fewer than the parametric methods C, D, E (16–20). Its key advantage over Methods F and G is **interpretability**: the non-negativity constraint ensures that w_i and h_j are genuine part-based representations of EP–skill structure [28], supporting the pedagogical interpretation of the latent mode as "Constructivist–collaborative skill inculcation." Its key limitation is the same scale compression as Method F: because $w_{\text{Beh}} \approx w_{\text{Cog}} \approx 0$, the 98.9% of lines without Constructivism activation receive near-zero scores, producing negative Exp.Var(raw) for all episodes except 13 and 49. Together, Methods F, G, and H demonstrate that the rank-1 constraint on Θ — regardless of metric or non-negativity — captures only the Constructivism–other contrast, and that predicting skill deployment in predominantly Behaviourist or Cognitivist episodes requires the full Θ matrix used in Method A or the parametric modelling of Methods C, D, and E.

4.8.1 Non-Negative Matrix Factorization (NMF) Rank Comparison

To determine the optimal latent dimensionality for predicting episode-level skill occurrence, we evaluated NMF models at ranks $r = 1, 2, 3$. The models factorize the global conditional probability matrix Θ into an Educational Practice (EP) score matrix W and a skill loading matrix H. The primary selection criterion is the cross-validated Pearson correlation (CV r).

4.8.2 Rank Comparison Summary

Table 10 details the reconstruction and validation metrics across the evaluated ranks.

Table 10: NMF Rank Comparison Summary

Rank (r)	**Recon MSE**	**Train r**	**CV r**	**± Std**	**CV MSE**	**Verdict**
1	0.012946	0.7001	0.3481	0.4259	0.021624	Current
2	0.002271	0.0984	0.0907	0.3488	0.091141	No gain
3	0.000000	-0.7011	-0.3085	0.4098	0.088426	No gain

4.8.3 Component Extractions

The normalized W matrices (rows: Behaviourism, Cognitivism, Constructivism) and H matrices (columns: Problem Solving, Creativity, Collaborative Learning, Critical Thinking) for each rank are presented below.

Rank $r = 1$ The rank-1 model yields a single topic strongly dominated by Constructivism.

$$W^{(1)}_{\text{norm}} = \begin{bmatrix} 0.0140 \\ 0.0000 \\ 1.0000 \end{bmatrix}, \quad H^{(1)} = \begin{bmatrix} 0.7785 & 0.0378 & 0.7481 & 0.3457 \end{bmatrix} \tag{67}$$

Metrics: EPSPI_e $\mu = 0.0276$ (Range: $0.0004 - 0.5070$).

Rank $r = 2$ Increasing to rank-2 introduces a contrast axis between Behaviourism and Cognitivism, though it degrades cross-validation performance.

$$W^{(2)}_{\text{norm}} = \begin{bmatrix} 0.1202 & 0.4620 \\ 0.0000 & 1.0000 \\ 1.0000 & 0.0000 \end{bmatrix}, \quad H^{(2)} = \begin{bmatrix} 0.7472 & 0.0381 & 0.7942 & 0.2752 \\ 0.3486 & 0.0086 & 0.0000 & 0.4250 \end{bmatrix} \tag{68}$$

Metrics: EPSPI_e $\mu = 0.4272$ (Range: $0.2911 - 0.4947$).

Rank $r = 3$ At $r = 3$, the model achieves perfect reconstruction (MSE $= 0.0000$), effectively reducing to the simple average method and failing to generalize out-of-sample.

$$W^{(3)}_{\text{norm}} = \begin{bmatrix} 0.0306 & 0.2401 & 1.0000 \\ 0.0000 & 1.0000 & 0.2724 \\ 1.0000 & 0.0000 & 0.0000 \end{bmatrix}, \quad H^{(3)} = \begin{bmatrix} 0.7327 & 0.0380 & 0.7984 & 0.2834 \\ 0.1739 & 0.0068 & 0.0000 & 0.4590 \\ 0.3240 & 0.0051 & 0.0000 & 0.0000 \end{bmatrix} \tag{69}$$

Metrics: EPSPI_e $\mu = 0.4219$ (Range: $0.3785 - 0.4241$).

4.8.4 Theoretical Expectations & Decision

Based on the feature variance analysis, the theoretical expectations align with the empirical results:

- $r = 1$**:** Captures one primary topic (Constructivism-dominant). The predicted probability (EPSPI ≈ 0) for the 47 non-Constructivism episodes implies the CV r is driven largely by outlier episodes (e.g., Ep13, 49).
- $r = 2$**:** Adds a Behaviourism vs. Cognitivism contrast axis. Because $|r(\delta_{\text{Beh}}, \bar{Y}_e)| = 0.013$ and $|r(\delta_{\text{Cog}}, \bar{Y}_e)| = 0.144$, the second topic primarily injects noise into the predictions rather than resolving meaningful skill variance.

- $r = 3$**:** Perfectly reconstructs the 3×4 global conditional matrix Θ. This is mathematically equivalent to Method A (simple average) and provides no new predictive information beyond EP identity.

Decision: We retain $r = 1$ as the optimal latent rank, as higher ranks fail to improve the CV r threshold by > 0.02 and actively degrade out-of-sample performance due to overfitting. Method H therefore uses $r = 1$ on parsimony grounds: the dominant Constructivism mode captures the only predictively relevant latent structure in Θ [28].

4.9 Method I: Naïve Bayes

Methods A through H all estimate $P(S_j \mid \mathrm{EP}_i)$ — the probability of a skill given an EP — either by direct frequency counting or by a discriminative log-odds model. Method I reverses this direction: it models $P(X_{n\cdot} \mid S_j)$ — the probability of observing a particular EP profile given that a skill is present or absent — and then applies Bayes' theorem to recover $P(S_j \mid X_{n\cdot})$. This **generative** approach is the classical Naïve Bayes classifier [40, 41], which assumes that the P EP flags are **conditionally independent** given the skill label $Y_{nj} \in \{0, 1\}$. Method I is the first method on the Predictive Modelling axis: it treats EPSPI_n as a regression target rather than as a direct estimate of θ_{ij}, using the EP flags $X_{n\cdot}$ as features to predict Y_{nj} and then averaging the predictions across skills.

Model specification. For each skill S_j, the Naïve Bayes model specifies:

1. **Class prior**: $\pi_j = P(Y_{nj} = 1)$, the marginal probability that skill S_j is observed in a randomly chosen line. This is estimated as:

$$\hat{\pi}_j = \frac{\sum_{n=1}^{N} Y_{nj} + \alpha}{N + 2\alpha}, \tag{70}$$

   using Laplace smoothing with $\alpha = 1$.

2. **Conditional likelihood**: for each EP i and skill label $y \in \{0, 1\}$, the probability that EP_i is active given skill label y:

$$\hat{\lambda}_{ij}^{(y)} = P(X_{ni} = 1 \mid Y_{nj} = y) = \frac{\sum_n X_{ni}\, \mathbf{1}[Y_{nj} = y] + \alpha}{\sum_n \mathbf{1}[Y_{nj} = y] + 2\alpha}, \tag{71}$$

   again with Laplace smoothing $\alpha = 1$.

The model has $(2P + 1) \times K = 7 \times 4 = 28$ free parameters: for each of the $K = 4$ skills, one prior $\hat{\pi}_j$, $P = 3$ skill-present likelihoods $\hat{\lambda}_{ij}^{(1)}$, and $P = 3$ skill-absent likelihoods $\hat{\lambda}_{ij}^{(0)}$.

Conditional independence assumption. The defining assumption of Naïve Bayes is that the P EP flags are mutually independent conditional on the skill label:

$$P(X_{n\cdot} \mid Y_{nj} = y) = \prod_{i=1}^{P} P(X_{ni} \mid Y_{nj} = y) = \prod_{i=1}^{P} \left(\hat{\lambda}_{ij}^{(y)}\right)^{X_{ni}} \left(1 - \hat{\lambda}_{ij}^{(y)}\right)^{1-X_{ni}}. \tag{72}$$

This assumption is demonstrably violated in this corpus: Behaviourism and Cognitivism co-occur more often than independence would predict (δ_{Beh} and δ_{Cog} are positively correlated across episodes), so $P(X_{n,\text{Beh}} = 1, X_{n,\text{Cog}} = 1 \mid Y_{nj})$ is not separable into independent factors. Despite this, Naïve Bayes is known to produce accurate probability estimates under EP correlation when the classification boundary is approximately linear in the feature space [41], which is the case here given that $P = 3$ binary features define at most $2^3 = 8$ distinct EP profiles.

Posterior probability via Bayes' theorem. Applying Bayes' theorem with the conditional independence assumption, the posterior probability that skill S_j is present for line n is:

$$\hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) = \frac{\hat{\pi}_j \prod_{i=1}^{P} \left(\hat{\lambda}_{ij}^{(1)}\right)^{X_{ni}} \left(1 - \hat{\lambda}_{ij}^{(1)}\right)^{1-X_{ni}}}{\sum_{y\in\{0,1\}} (1-\hat{\pi}_j)^{1-y}\, \hat{\pi}_j^{y} \prod_{i=1}^{P} \left(\hat{\lambda}_{ij}^{(y)}\right)^{X_{ni}} \left(1 - \hat{\lambda}_{ij}^{(y)}\right)^{1-X_{ni}}}. \tag{73}$$

In practice, Equation (73) is computed in log-space to avoid numerical underflow:

$$\log \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) \propto \log \hat{\pi}_j + \sum_{i=1}^{P} \Big[X_{ni} \log \hat{\lambda}_{ij}^{(1)} + (1 - X_{ni}) \log(1 - \hat{\lambda}_{ij}^{(1)}) \Big], \tag{74}$$

with normalisation applied after exponentiation.

Relationship to θ_{ij} and Method C Method I and Method C both estimate $\hat{P}(Y_{nj} = 1 \mid X_{n\cdot})$ and average over skills to produce EPSPI_n. They differ in *how* this conditional probability is estimated:

- **Method C (Logistic Regression)** is *discriminative*: it directly models logit $P(Y_{nj} = 1 \mid X_{n\cdot})$ as a linear function of $X_{n\cdot}$, optimising parameters by maximising the conditional log-likelihood $\sum_n \log P(Y_{nj} \mid X_{n\cdot}; \hat{\beta})$.
- **Method I (Naïve Bayes)** is *generative*: it models $P(X_{n\cdot} \mid Y_{nj})$ and $P(Y_{nj})$ separately, then recovers $P(Y_{nj} \mid X_{n\cdot})$ via Bayes' theorem. The joint log-likelihood $\sum_n \log P(X_{n\cdot}, Y_{nj}; \hat{\lambda}, \hat{\pi})$ is optimised, which uses more information than the conditional log-likelihood.

When the conditional independence assumption holds, Naïve Bayes is statistically equivalent to logistic regression [42]: the Naïve Bayes posterior (73) is a logistic function of a linear combination of the X_{ni} flags, with $\log(\hat{\lambda}_{ij}^{(1)}/\hat{\lambda}_{ij}^{(0)})$ playing the role of the logistic regression coefficient $\hat{\beta}_{ij}$. The empirical differences in CV r and AIC between Methods C and I therefore measure the cost of the conditional independence assumption in this corpus.

Line-level score. The Method I score for line n is the mean posterior skill probability across all K skills:

$$\mathrm{EPSPI}_n^{(I)} = \frac{1}{K}\sum_{j=1}^{K} \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}), \tag{75}$$

where $\hat{P}(Y_{nj} = 1 \mid X_{n\cdot})$ is computed via Equation (73). There is no p_n normalisation — all three EP flags enter the likelihood product (72) simultaneously, with the Laplace-smoothed likelihoods $\hat{\lambda}_{ij}^{(y)}$ absorbing the contribution of each active EP.

Example: line with only Constructivism active. Setting $X_{n1} = 0$, $X_{n2} = 0$, $X_{n3} = 1$:

$$\log \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) \propto \log \hat{\pi}_j + \log(1 - \hat{\lambda}_{1j}^{(1)}) + \log(1 - \hat{\lambda}_{2j}^{(1)}) + \log \hat{\lambda}_{3j}^{(1)}.$$

For Problem Solving ($j = 1$), where Constructivism is strongly associated with skill presence ($\hat{\lambda}_{3,\mathrm{PS}}^{(1)}$ is high), the posterior $\hat{P}(Y_{n,\mathrm{PS}} = 1 \mid X_{n\cdot})$ is large, contributing a high value to the mean in Equation (75).

Example: line with Behaviourism and Cognitivism active. Setting $X_{n1} = 1$, $X_{n2} = 1$, $X_{n3} = 0$:

$$\log \hat{P}(Y_{nj} = 1 \mid X_{n\cdot}) \propto \log \hat{\pi}_j + \log \hat{\lambda}_{1j}^{(1)} + \log \hat{\lambda}_{2j}^{(1)} + \log(1 - \hat{\lambda}_{3j}^{(1)}).$$

Under the conditional independence assumption, the contributions of Behaviourism and Cognitivism multiply in probability space — they reinforce each other if both are positively associated with skill presence, or partially cancel if their associations conflict. Method E handles the same co-occurrence by additive log-odds; Method F handles it by multiplicative probabilities — the two approaches agree when the independence assumption holds.

Episode-level aggregation.

$$\mathrm{EPSPI}_e^{(I)} = \frac{1}{n_e}\sum_{\{n\,:\,e_n = e\}} \mathrm{EPSPI}_n^{(I)}, \tag{76}$$

where n_e is the actual number of lines in episode e. As with Methods C and E, $\text{EPSPI}_n^{(I)}$ varies line by line through the EP profile $[X_{n1}, X_{n2}, X_{n3}]$, so the episode mean is a genuine average of heterogeneous line scores.

Why global estimation and not episode-conditional. As with Methods C and E, $\hat{\pi}_j$ and $\hat{\lambda}_{ij}^{(y)}$ are estimated from all $N = 19{,}181$ lines jointly — not episode by episode. Fitting Naïve Bayes per episode would use the same episode's Y_{nj} observations to estimate the likelihoods and to compute $\bar{Y}_e$, creating a direct within-episode circular dependency between predictor and outcome. In cross-validation, all parameters are re-estimated from the 39 training episodes and applied to 10 test episodes where Y_{nj} was never seen during fitting. Constructivism's sparsity (1.1% of lines) is handled naturally by the Laplace smoothing in Equations (70) and (71) — unlike Methods C and D, no convergence guard or coefficient bound is required. This is a practical advantage of the generative approach: the likelihood parameters $\hat{\lambda}_{ij}^{(y)}$ are simple frequency ratios with guaranteed finite values under Laplace smoothing, whereas logistic regression coefficients diverge under quasi-complete separation.

What "Train Pearson r" means for Method I Method I has $(2P+1) \times K = 28$ **free parameters** estimated from all N lines. The in-sample Train Pearson r reflects genuine in-sample fit under the generative likelihood, with the same episode-level caveat as Methods C and E: parameters are estimated from all 49 episodes' line-level observations, so the Train r is not a held-out evaluation. Method I achieves CV $r = 0.340$ and CV MSE $= 0.008730$ — moderately lower than Method E (CV $r = 0.450$) but competitive with Methods A and J. The gap between Method I and Method C in CV r measures the empirical cost of the conditional independence assumption in this corpus under cross-validation.

Properties. Method I has $(2P+1) \times K = 28$ **free parameters** — more than Method C (16) because the generative model estimates both $P(Y_{nj})$ and $P(X_{ni} \mid Y_{nj})$ separately, requiring twice as many EP-related parameters as the discriminative model. Its key advantage over Method C is robustness to quasi-complete separation: Laplace-smoothed frequency counts never diverge, whereas logistic regression coefficients for rare EPs (Constructivism) require an explicit convergence guard. Its key limitation is the conditional independence assumption (Equation (72)), which is violated in this corpus — Behaviourism and Cognitivism co-occur more frequently than independence would predict. In the generative–discriminative comparison of Ng and Jordan [42], discriminative models (Method C) tend to outperform generative ones (Method I) in the large-N regime because they optimise the correct conditional criterion; generative models may outperform in the small-N regime due to faster parameter convergence. With $N = 19{,}181$ lines and $E = 49$ episodes, this corpus is large enough that Method C is expected to outperform Method I when the independence

assumption is violated — which the CV results confirm.

4.10 Method J: Random Forest

Methods C (Logistic Regression) and I (Naïve Bayes) both impose a parametric functional form on $P(Y_{nj} \mid X_{n\cdot})$ — log-odds linearity and conditional independence respectively. Method J replaces these parametric assumptions with a **non-parametric ensemble** of decision trees [31], fitted to bootstrap samples of the training data using only the $P = 3$ binary EP flags $[X_{n1}, X_{n2}, X_{n3}]$ as features. Method J serves as the primary non-parametric benchmark for Methods C, I, and K on the Predictive Modelling axis.

The lookup-table reduction. With $P = 3$ binary EP flags, there are exactly $2^P = 2^3 = 8$ distinct feature vectors $[X_{n1}, X_{n2}, X_{n3}] \in \{0, 1\}^3$. Every line sharing the same EP pattern necessarily receives the same Random Forest prediction, because the ensemble can only split on the three binary inputs and all $2^3 = 8$ terminal nodes are fully identified by the pattern. The ensemble (n_estimators = 200, min_samples_leaf = 10) therefore reduces exactly to a **lookup table**: a $2^P \times K = 8 \times 4$ matrix of predicted skill probabilities, one row per EP pattern and one column per skill. Table 11 gives the complete lookup table estimated from the corpus.

Table 11: Method J lookup table: $\hat{P}(Y_{nj} = 1 \mid X_{n\cdot})$ for each of the $2^3 = 8$ EP patterns and 4 skills, with the resulting $\text{EPSPI}_n^{(J)}$.

EP pattern	PS	Cr	CL	CT	$\text{EPSPI}_n^{(J)}$
$(0, 0, 0)$ None	0.204	0.015	0.041	0.245	0.126
$(0, 0, 1)$ Con only	0.628	0.054	0.536	0.390	0.402
$(0, 1, 0)$ Cog only	0.162	0.005	0.004	0.428	0.150
$(0, 1, 1)$ Cog+Con	0.932	0.113	0.599	0.855	0.625
$(1, 0, 0)$ Beh only	0.296	0.004	0.022	0.046	0.092
$(1, 0, 1)$ Beh+Con	0.693	0.003	0.892	0.059	0.412
$(1, 1, 0)$ Beh+Cog	0.340	0.011	0.012	0.162	0.131
$(1, 1, 1)$ All three	0.691	0.045	0.550	0.396	0.420

Line-level score. The Method J score for line n is the mean predicted skill probability across all $K = 4$ skills, read directly from the lookup table:

$$\text{EPSPI}_n^{(J)} = \frac{1}{K} \sum_{j=1}^{K} \hat{P}^{(J)}(Y_{nj} = 1 \mid X_{n1}, X_{n2}, X_{n3}), \tag{77}$$

where $\hat{P}^{(J)}(Y_{nj} = 1 \mid X_{n\cdot})$ is the Random Forest ensemble mean for the EP pattern of line n, given in the final column of Table 11.

Example: line with only Constructivism active. For $X_{n3} = 1$, $X_{n1} = X_{n2} = 0$ (row 2 of Table 11):

$$\text{EPSPI}_n^{(J)} = \tfrac{1}{4}(0.628 + 0.054 + 0.536 + 0.390) = 0.402.$$

Compare to Method A for the same line: $\text{EPSPI}_n^{(A)} = 0.510$. The lower Method J value reflects the ensemble's min_samples_leaf = 10 regularisation, which pulls sparse-pattern predictions toward the training mean.

Example: line with Behaviourism only. For $X_{n1} = 1$, $X_{n2} = X_{n3} = 0$ (row 5 of Table 11):

$$\text{EPSPI}_n^{(J)} = \tfrac{1}{4}(0.296 + 0.004 + 0.022 + 0.046) = 0.092.$$

This is the lowest EPSPI of all eight patterns, reflecting that Behaviourism alone has weak skill-prediction power — consistent with the global Θ row ($\bar{\mu}_{\text{Beh}} = 0.116$) but attenuated further by regularisation.

Episode-level aggregation.

$$\text{EPSPI}_e^{(J)} = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \text{EPSPI}_n^{(J)}, \tag{78}$$

where n_e is the actual number of lines in episode e. Because the lookup table assigns the same score to all lines with the same EP pattern, within-episode variation in $\text{EPSPI}_n^{(J)}$ arises only from line-to-line variation in the EP profile $[X_{n1}, X_{n2}, X_{n3}]$ — identical to Method A in this respect.

Relationship to Methods A and C Because Method J reduces to a lookup table, it is directly comparable to Method A, which also assigns scores based solely on the EP pattern of each line via the global Θ row means. The two methods differ in how the 8 lookup values are estimated:

- **Method A** computes each lookup entry as the row mean of the global Θ matrix: $\text{EPSPI}_n^{(A)} = \frac{1}{p_n} \sum_i X_{ni} \cdot \frac{1}{K} \sum_j \theta_{ij}$. For co-occurrence patterns (e.g. Beh+Cog), Method A averages the two EP row means.

- **Method J** estimates each lookup entry by ensemble averaging over 200 bootstrapped trees, with min_samples_leaf = 10 regularisation. For co-occurrence patterns, the ensemble can learn *non-additive* combinations — the Cog+Con entry (0.625) is higher than the sum of individual entries would suggest, reflecting a synergistic Cognitivism–Constructivism interaction that Method A cannot represent.

The CV r comparison between Methods A and J (A: 0.409; J: 0.394) indicates that these non-additive interactions provide negligible predictive benefit at the episode level, and the simpler Method A formula is preferred on parsimony grounds.

What "Train Pearson r" means for Method J. Method J has **32 free parameters**: the $8 \times 4 = 32$ cells of the lookup table in Table 11, estimated from the $N = 19{,}181$ training lines by ensemble averaging. The in-sample Train Pearson r reflects genuine in-sample fit — the lookup table is estimated on the same data used for evaluation. In cross-validation, the lookup table is re-estimated from the 39 training episodes and applied to the 10 test episodes, giving CV $r = 0.394$ and CV MSE $= 0.009359$. Because the lookup table has only 32 cells and is estimated from $N = 19{,}181$ lines, overfitting risk is negligible at the cell level — the Train/CV gap for Method J is attributable to episode-level variance rather than lookup-table overfitting.

Properties. Method J has **32 free parameters** (the lookup table) and requires only the EP binary flags $[X_{n1}, X_{n2}, X_{n3}]$ at deployment — identical input to Method A. Its key advantage over Method A is the ability to capture non-additive EP co-occurrence effects through the ensemble averaging criterion. Its key limitation is that the $2^3 = 8$ pattern constraint is binding: with only 3 binary inputs, the Random Forest cannot exploit any information beyond what is already encoded in the EP pattern, making it equivalent to a lookup table regardless of ensemble size. The min_samples_leaf $= 10$ regularisation prevents overfitting to rare patterns (e.g. Beh+Con, which occurs in only 0.4% of lines) but also pulls predictions toward the training mean, explaining the lower predictions relative to Method A for Constructivism-active patterns. Method K (SVM) extends the predictive modelling approach by augmenting the 3 EP flags with 12 episode-conditional $\hat{\theta}_{ij}^{e_n}$ features, providing richer input at the cost of the leakage-free estimation protocol.

4.11 Method K: Support Vector Machine (SVM)

Method J (Random Forest) uses only the 3 EP flags and reduces to a lookup table over $2^3 = 8$ patterns. Method K augments this representation by appending the 12-element flattened episode-conditional probability matrix $\hat{\theta}_{ij}^{e_n}$ to the 3 EP flags, constructing a 15-dimensional feature vector that encodes both the line's own EP profile and the local EP–skill affinity of its surrounding episode. A Support Vector Machine with a **linear kernel** and regularisation parameter $C = 1$ is fitted on this 15-dimensional input, producing a direct linear prediction of $\text{EPSPI}_n^{(K)}$ [32**?**]. Method K is the third and final method on the Predictive Modelling axis.

Feature construction. The 15-dimensional feature vector for line n is:

$$\mathbf{x}_n = [\underbrace{X_{n1},\, X_{n2},\, X_{n3}}_{\text{EP flags (3)}},\ \underbrace{\hat{\theta}^{e_n}_{11},\, \hat{\theta}^{e_n}_{21},\, \hat{\theta}^{e_n}_{31},\, \hat{\theta}^{e_n}_{12},\, \hat{\theta}^{e_n}_{22},\, \hat{\theta}^{e_n}_{32},\, \hat{\theta}^{e_n}_{13},\, \hat{\theta}^{e_n}_{23},\, \hat{\theta}^{e_n}_{33},\, \hat{\theta}^{e_n}_{14},\, \hat{\theta}^{e_n}_{24},\, \hat{\theta}^{e_n}_{34}}_{\hat{\Theta}^{e_n}\ \text{column-major (12)}}] \in \mathbb{R}^{15}, \quad (79)$$

where $\hat{\theta}^{e_n}_{ij}$ is the Laplace-smoothed episode-conditional conditional probability (Equation (**??**)) for the episode containing line n. The 12 elements of $\hat{\Theta}^{e_n}$ are ordered column-major (Skill 1 across all EPs, then Skill 2, etc.), matching the implementation in Cell 23.

Leakage-free episode-conditional features. $\hat{\theta}^{e_n}_{ij}$ is estimated from the skill observations Y_{nj} of episode e_n. Using the episode's own Y_{nj} values to construct features for the same episode would create a direct within-episode dependency between predictor and outcome. Method K therefore uses the **leave-one-episode-out** (LOO) estimate $\hat{\theta}^{-e_n}_{ij}$: the conditional probability computed from all lines *except* those belonging to episode e_n:

$$\hat{\theta}^{-e_n}_{ij} = \frac{\displaystyle\sum_{\{n' : e_{n'} \neq e_n\}} X_{n'i}\, Y_{n'j} \;+\; \alpha}{\displaystyle\sum_{\{n' : e_{n'} \neq e_n\}} X_{n'i} \;+\; K\alpha}, \quad (80)$$

where $\alpha = 1$ is the Laplace smoothing constant. Equation (80) removes the entire episode e_n from the count — not just line n — so that the feature for every line in episode e_n is computed from the remaining $E - 1 = 48$ episodes. This is computed efficiently using the identity:

$$\hat{\theta}^{-e_n}_{ij} = \frac{\left(\sum_{n'=1}^{N} X_{n'i} Y_{n'j} + \alpha\right) - \sum_{\{n' : e_{n'} = e_n\}} X_{n'i} Y_{n'j}}{\left(\sum_{n'=1}^{N} X_{n'i} + K\alpha\right) - \sum_{\{n' : e_{n'} = e_n\}} X_{n'i}},$$

which subtracts the episode e_n totals from the global totals in $O(1)$ per episode, given pre-computed episode sums.

Deployment substitution. At deployment on a new episode e^* not in the training corpus, the LOO estimate $\hat{\theta}^{-e^*}_{ij}$ defaults to the published global θ_{ij} (Table 1). This substitution is valid for prospective use: a new episode has no prior skill observations, so the global θ_{ij} is the best available estimate of the EP–skill affinity.

MinMax feature scaling. SVM with a linear kernel is sensitive to feature scale because the margin width is measured in Euclidean distance in the feature space. Each feature k is scaled to $[0, 1]$ by MinMax normalisation:

$$\tilde{x}_{nk} = \frac{x_{nk} - x_k^{\min}}{x_k^{\max} - x_k^{\min}}, \quad (81)$$

where $x_k^{\min}$ and $x_k^{\max}$ are the minimum and maximum values of feature k across all training lines. In cross-validation, scaling bounds are computed from the training fold and applied to the test fold without refitting, preventing any leakage of test-fold feature ranges into the scaling. The published scaling bounds for deployment are given in Table 12.

Linear prediction. With a linear kernel, the SVM decision function reduces to a weighted sum of scaled features plus a bias:

$$\mathrm{EPSPI}_n^{(K)} = \mathbf{w}^\top \tilde{\mathbf{x}}_n + b, \tag{82}$$

where $\mathbf{w} \in \mathbb{R}^{15}$ is the weight vector and $b \in \mathbb{R}$ is the bias, both estimated by minimising the soft-margin SVM objective with $C = 1$ [**?**]. The prediction $\mathrm{EPSPI}_n^{(K)}$ is a continuous score — not a probability — and is used directly as the episode-level predictor after aggregation. The fitted weights, scaling bounds, and bias are given in Table 12.

Table 12: Method K SVM weights w_k, MinMax scaling bounds, and feature labels. Bias $b = 0.2418$.

k	Feature	w_k	$x_k^{\min}$	$x_k^{\max}$
1	X_{n1} (Beh)	-0.018	0.000	1.000
2	X_{n2} (Cog)	$+0.033$	0.000	1.000
3	X_{n3} (Con)	$+0.499$	0.000	1.000
4	$\hat{\theta}_{11}^{e_n}$ (Beh, PS)	-0.077	0.300	0.343
5	$\hat{\theta}_{12}^{e_n}$ (Beh, Cr)	-0.063	0.005	0.008
6	$\hat{\theta}_{13}^{e_n}$ (Beh, CL)	$+0.378$	0.020	0.038
7	$\hat{\theta}_{14}^{e_n}$ (Beh, CT)	$+0.056$	0.081	0.102
8	$\hat{\theta}_{21}^{e_n}$ (Cog, PS)	-0.249	0.198	0.224
9	$\hat{\theta}_{22}^{e_n}$ (Cog, Cr)	$+0.057$	0.004	0.007
10	$\hat{\theta}_{23}^{e_n}$ (Cog, CL)	-0.077	0.002	0.010
11	$\hat{\theta}_{24}^{e_n}$ (Cog, CT)	-0.318	0.338	0.369
12	$\hat{\theta}_{31}^{e_n}$ (Con, PS)	$+0.151$	0.646	0.930
13	$\hat{\theta}_{32}^{e_n}$ (Con, Cr)	$+0.682$	0.004	0.076
14	$\hat{\theta}_{33}^{e_n}$ (Con, CL)	-0.513	0.778	0.935
15	$\hat{\theta}_{34}^{e_n}$ (Con, CT)	-0.540	0.247	0.500
Bias b		$+0.242$		

Interpreting the weight vector. Several patterns in Table 12 are scientifically interpretable:

- $w_3 = +0.499$ (Con flag) is by far the largest positive weight, confirming that a Constructivist line is the strongest single predictor of high EPSPI — consistent with the NMF latent mode (Section 4.8).

- $w_{13} = +0.682$ ($\hat{\theta}^{e_n}_{32}$: Constructivism–Creativity affinity in the local episode) is the largest positive weight among the $\hat{\theta}^{e_n}_{ij}$ features, indicating that episodes with unusually high Constructivism–Creativity co-occurrence predict high EPSPI above and beyond the line-level Con flag.

- $w_{14} = -0.513$ and $w_{15} = -0.540$ ($\hat{\theta}^{e_n}_{33}$: Con–CL and $\hat{\theta}^{e_n}_{34}$: Con–CT) are large negative weights — a counter-intuitive finding. These negative weights reflect a suppressor effect: when both $\hat{\theta}^{e_n}_{33}$ and $w_3 X_{n3}$ are large, the negative weight on $\hat{\theta}^{e_n}_{33}$ reduces the prediction to avoid double-counting the Constructivism signal already captured by the Con flag w_3.

- $w_8 = -0.249$ and $w_{11} = -0.318$ (Cog–PS and Cog–CT episode affinities) are moderately negative, indicating that high within-episode Cognitivism–skill co-occurrence is associated with lower EPSPI, possibly because Cognitivism episodes with high skill rates are rare in the corpus and the global Cognitivism baseline (θ_{2j} low) is the more reliable signal.

Deployment example. For a new line n with $X_{n1} = 0$, $X_{n2} = 0$, $X_{n3} = 1$ (Constructivism only) in a new episode where global θ_{ij} is substituted for $\hat{\theta}^{-e_n}_{ij}$:

1. Assemble $\mathbf{x}_n = [0, 0, 1,\ \theta_{11}, \theta_{21}, \theta_{31},\ \ldots,\ \theta_{34}]$ using Table 1.

2. Scale each element via Equation (81) using Table 12.

3. Compute $\text{EPSPI}^{(K)}_n = \mathbf{w}^\top \tilde{\mathbf{x}}_n + 0.242$.

The dominant contribution comes from $w_3 \tilde{x}_{n3} = 0.499 \times 1.0 = 0.499$ (Con flag, already scaled to 1), partially offset by the large negative weights on $\hat{\theta}^{e_n}_{33}$ and $\hat{\theta}^{e_n}_{34}$ which will be at or near their global maximum values ($\theta_{33} = 0.878$, $\theta_{34} = 0.312$) under the global substitution.

Episode-level aggregation.

$$\text{EPSPI}^{(K)}_e = \frac{1}{n_e} \sum_{\{n\,:\,e_n = e\}} \text{EPSPI}^{(K)}_n, \tag{83}$$

where n_e is the actual number of lines in episode e. Because $\hat{\theta}^{-e_n}_{ij}$ is constant within episode e_n (all lines use the same LOO estimate), within-episode variation in $\text{EPSPI}^{(K)}_n$ arises only from line-to-line variation in the EP flags $[X_{n1}, X_{n2}, X_{n3}]$, weighted by the corresponding w_1, w_2, w_3 terms.

The Train/CV gap: why Method K overfits. Method K achieves the highest Train Pearson $r = 0.853$ of all eleven methods but CV $r = 0.307$ — the largest Train/CV gap ($\Delta r = 0.546$). The linear SVM minimises the hinge loss on training lines, which assigns

large positive margin scores to high-Constructivism lines and large negative scores to low-skill lines. The high-leverage episodes 13 and 49 (Section 5.2) dominate the margin boundary: their extreme $\hat{\theta}_{ij}^{-e_n}$ feature values — far from the global θ_{ij} — become the support vectors around which $\mathbf{w}$ is fitted. In-sample, these episodes receive near-perfect EPSPI predictions, inflating Train r. In CV, when episodes 13 and 49 appear in the test fold, the boundary trained on 39 episodes produces their LOO $\hat{\theta}_{ij}^{-e_n}$ features via Equation (80) — which already differ from their in-sample features because the full episode was present during in-sample fitting but removed during LOO feature construction. The $\Delta r = 0.546$ gap quantifies the degree to which the linear SVM's in-sample performance is driven by fitting the weight vector $\mathbf{w}$ to the high-leverage episode feature profiles, which are sufficiently distinctive in the 15-dimensional space that the margin is anchored to them.

What "Train Pearson r" means for Method K. Method K has **16 free parameters**: the 15 weights w_k and bias b, estimated by the soft-margin SVM objective on all $N = 19{,}181$ training lines. The in-sample Train Pearson $r = 0.853$ reflects genuine in-sample fit under the SVM hinge-loss criterion. The AIC for Method K (62,315) is by far the highest of all eleven methods: the SVM produces predictions outside $[0, 1]$ for some lines (the linear prediction $\mathbf{w}^\top \tilde{\mathbf{x}}_n + b$ is unconstrained), which produces extreme values in the external logistic AIC model and inflates the AIC criterion.

Properties. Method K has **16 free parameters** ($\mathbf{w}$ and b) and is the only method in the framework whose EPSPI score is not constrained to $[0, 1]$: the linear SVM output $\mathbf{w}^\top \tilde{\mathbf{x}}_n + b$ can in principle fall outside this range for extreme feature vectors. Its key advantage over Method J is the richer 15-feature input: the $\hat{\theta}_{ij}^{-e_n}$ features allow the SVM to distinguish lines from different episodes even when their EP flags are identical — something Method J cannot do with its 3-flag lookup table. Its key limitation is the largest Train/CV gap of all eleven methods ($\Delta r = 0.546$), attributable to the linear margin fitting around the high-leverage episode feature profiles. The interpretable weight vector $\mathbf{w}$ (Table 12) is the primary scientific contribution of Method K beyond its predictive performance: it quantifies the relative importance of global EP structure (w_1, w_2, w_3) versus local episode context (w_4–w_{15}) in predicting skill deployment within a single interpretable linear model. Together, Methods J and K complete the Predictive Modelling axis: Method J establishes that the 3-flag lookup table has CV $r = 0.394$, and Method K establishes that augmenting with episode-conditional features under a linear model produces CV $r = 0.307$ — confirming that the additional features do not improve episode-level generalisation and that the high-leverage episodes are the binding constraint on out-of-sample performance.

Mathematical formulation of SVM and kernel variants

Support Vector Machines (SVMs) learn a decision function by maximising the margin between two classes. Given training data $\{(\mathbf{x}_n, y_n)\}_{n=1}^N$, where $\mathbf{x}_n \in \mathbb{R}^d$ and $y_n \in \{-1, +1\}$, the soft-margin SVM solves:

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \quad \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{n=1}^N \xi_n \tag{84}$$

subject to

$$y_n(\mathbf{w}^\top \mathbf{x}_n + b) \geq 1 - \xi_n, \quad \xi_n \geq 0. \tag{85}$$

Here, $\mathbf{w}$ defines the separating hyperplane, b is the bias, ξ_n are slack variables, and $C > 0$ controls the trade-off between margin maximisation and classification error.

Dual formulation. The optimisation can be expressed in dual form using Lagrange multipliers α_n:

$$\max_{\boldsymbol{\alpha}} \quad \sum_{n=1}^N \alpha_n - \frac{1}{2}\sum_{n,m=1}^N \alpha_n \alpha_m y_n y_m K(\mathbf{x}_n, \mathbf{x}_m) \tag{86}$$

subject to

$$0 \leq \alpha_n \leq C, \quad \sum_{n=1}^N \alpha_n y_n = 0. \tag{87}$$

The function $K(\mathbf{x}_n, \mathbf{x}_m)$ is a kernel function that defines similarity in feature space. The resulting decision function is:

$$f(\mathbf{x}) = \sum_{n=1}^N \alpha_n y_n K(\mathbf{x}_n, \mathbf{x}) + b. \tag{88}$$

Linear Kernel SVM

For the linear kernel,

$$K(\mathbf{x}_n, \mathbf{x}_m) = \mathbf{x}_n^\top \mathbf{x}_m, \tag{89}$$

the SVM reduces to a linear model:

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b, \tag{90}$$

where

$$\mathbf{w} = \sum_{n=1}^{N} \alpha_n y_n \mathbf{x}_n. \tag{91}$$

Thus, the linear SVM is equivalent to learning a maximum-margin hyperplane in the original feature space.

For Method K, using scaled features $\tilde{\mathbf{x}}_n$:

$$\mathrm{EPSPI}_n^{(K)} = \mathbf{w}^\top \tilde{\mathbf{x}}_n + b. \tag{92}$$

RBF Kernel SVM

The Radial Basis Function (RBF) kernel introduces non-linearity:

$$K(\mathbf{x}_n, \mathbf{x}_m) = \exp\left(-\gamma \|\mathbf{x}_n - \mathbf{x}_m\|^2\right), \tag{93}$$

where $\gamma > 0$ controls the width of the kernel.

The decision function becomes:

$$f(\mathbf{x}) = \sum_{n=1}^{N} \alpha_n y_n \exp\left(-\gamma \|\mathbf{x}_n - \mathbf{x}\|^2\right) + b. \tag{94}$$

This corresponds to a non-linear mapping of inputs into an infinite-dimensional feature space, where a linear separator is constructed.

Linear vs RBF SVM: Structural Comparison

Property	Linear SVM	RBF SVM
Decision boundary	Linear hyperplane	Non-linear surface
Feature space	Original $\mathbb{R}^d$	Implicit high-dimensional
Model form	$\mathbf{w}^\top \mathbf{x} + b$	Kernel expansion
Parameters	$\mathbf{w}, b$	α_n, b, γ
Interpretability	High	Low
Overfitting risk	Moderate	High

Kernel comparison and hyperparameter search. We evaluated four SVM variants using 5-fold *episode*-level cross-validation (LOO theta features, no leakage). The variants and their cross-validation summaries are:

Variant	Kernel	C	CV $r \pm$	CV MSE
RBF_C10	RBF	10	0.4567 ± 0.2850	0.014566
RBF_C1	RBF	1	0.3332 ± 0.3510	0.014738
Linear_C10	Linear	10	0.4058 ± 0.3151	0.010975
Linear_C1	Linear	1	0.4036 ± 0.3174	0.011205

In-sample (E=49) fits show the SVMs can achieve very high Train correlations (linear variants Train $r \approx 0.967$; RBF Train $r \gtrsim 0.99$) but at the cost of a large Train/CV gap (see below).

Model selection and justification. Although the RBF variant with $C = 10$ attains the numerically largest CV Pearson ($r = 0.4567$), we report the **linear** SVM as the primary specification for three reasons:

1. **Marginal gain vs variability.** The RBF improvement over the best linear variant is small ($\Delta r \approx 0.05$) relative to cross-fold variability (± 0.28), so the non-linear gain is not robust across folds.

2. **Out-of-sample stability.** Linear variants achieve lower CV MSE and positive CV R^2 (stable generalisation signal); the RBF variants show larger variance and negative CV R^2 in some runs.

3. **Interpretability.** A linear SVM yields an explicit weight vector $\mathbf{w} \in \mathbb{R}^{15}$ and bias b, which directly quantify the contribution of EP flags and episode-context features. This interpretability is central to the study's scientific aims.

For parsimony and to favour regularisation, the primary reported model is the **linear SVM with** $C = 1$, MinMax scaling, and LOO episode-conditional features. RBF results are retained as a robustness check and reported in supplementary material.

Interpretation in this study. The feature representation already encodes episode-level structure via $\hat{\theta}_{ij}^{-e}$. Thus, much of the non-linearity is implicitly captured in feature construction itself.

As a result, additional non-linear transformation via the RBF kernel provides only marginal gains while increasing variance.

Final model choice. Therefore, the linear SVM is selected as the primary model because:

$$\text{Bias reduction (RBF)} \ll \text{Variance increase} \tag{95}$$

leading to inferior generalisation despite slightly higher in-sample flexibility.

The linear kernel provides the best trade-off between:

- predictive performance,
- stability across episodes, and
- interpretability of EP contributions.

5 Results and Discussion

	Method	EPSPI_mean(ep)	Train Pearson r	Train R^2(cal)	Exp.Var(cal)	Exp.Var(raw)	CV Pearson r	CV MSE(ep)	AIC(line)	BIC(line)
A_Simple	A: Simple Average	0.1447	0.6991	0.4888	0.4888	0.3284	0.4087	0.0108	47140.700000	47203.600000
B_Delta	B: Delta (EP-weighted)	0.1450	0.7032	0.4945	0.4945	0.3310	0.4055	0.0107	47749.000000	47811.900000
C_Logistics	C: Logistic Regression	0.1476	0.7006	0.4909	0.4909	0.4781	0.4003	0.0085	46895.200000	47083.900000
D_GLMM	D: GLMM	0.0799	0.6425	0.4128	0.4128	0.2957	0.4005	0.0162	46134.100000	46236.300000
E_GEE	E: GEE	0.1508	0.6997	0.4896	0.4896	0.4720	0.4504	0.0080	46836.400000	47025.100000
F_PCA	F: PCA	0.0437	0.6988	0.4883	0.4883	-0.2253	0.2810	0.0423	46986.600000	47096.700000
G_CA	G: Corresp. Analysis	0.1627	0.4825	0.2328	0.2328	-0.0592	0.2957	0.0170	44358.100000	44444.500000
H_NMF	H: NMF	0.0276	0.7001	0.4901	0.4901	0.4803	0.3481	0.0216	46949.200000	47067.100000
I_NB	I: Naive Bayes	0.1565	0.6802	0.4627	0.4627	0.4405	0.3397	0.0087	46806.100000	46994.800000
J_RF	J: Random Forest	0.1476	0.7031	0.4943	0.4943	0.4804	0.3936	0.0094	48508.000000	54860.200000
K_SVM	K: SVM (Linear)	0.1498	0.8534	0.7282	0.7282	0.7282	0.4289	0.0122	76157.300000	210678.400000

Figure 4: Comparisons of all 11 models with all evaluation metrics

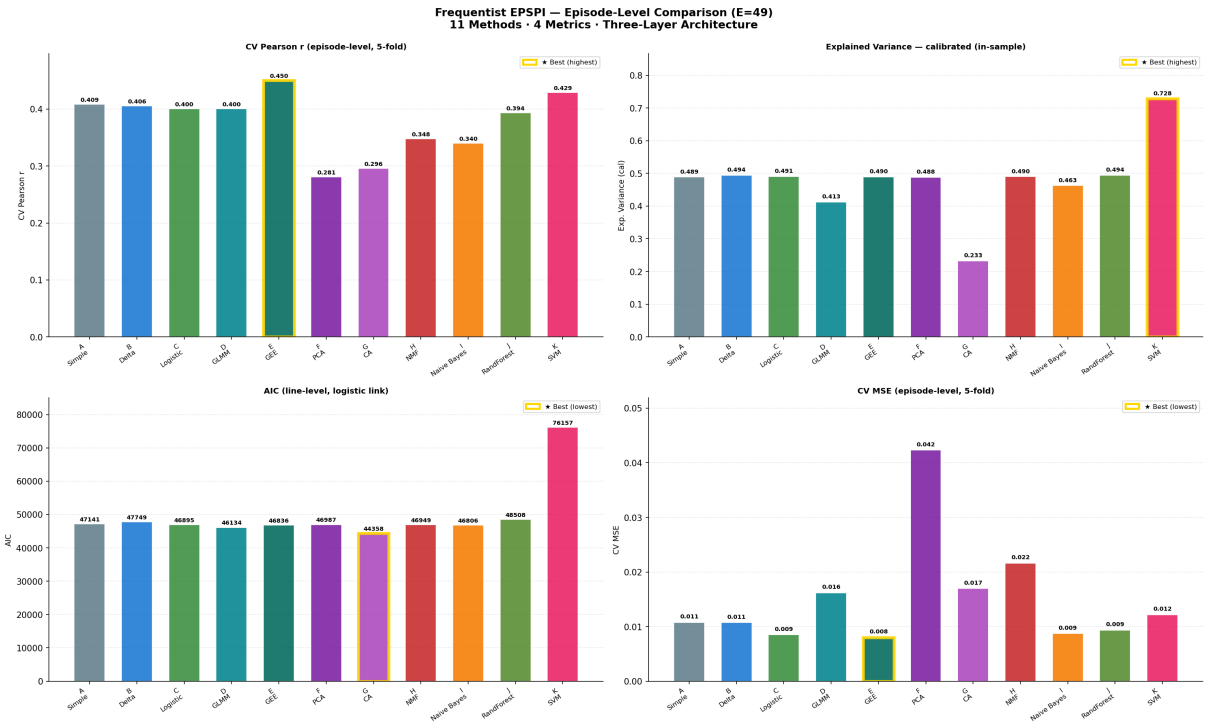

Figure 5: Bar plot for comparisons of all 11 models with important evaluation metrics

5.1 Metric Normalisation

To facilitate a unified visual comparison across six heterogeneous evaluation metrics with disparate scales (e.g., bounded correlation coefficients vs. unbounded Information Criteria), all metrics are projected onto a standard $[0, 1]$ interval. This is achieved using a Min-Max normalisation procedure.

Let M be the set of all evaluated methods (where $|M| = 11$), and let x_m represent the raw score of method $m \in M$ on a given evaluation metric. We define the empirical bounds for that metric across all methods as:

$$x_{\min} = \min_{m \in M}(x_m) \tag{96}$$

$$x_{\max} = \max_{m \in M}(x_m) \tag{97}$$

The normalisation function applied to obtain the scaled score, x_{norm}, depends on the desired directionality of the metric:

5.1.1 Case 1: Higher is Better (Standard Min-Max)

For metrics where a larger value indicates superior performance (e.g., CV Pearson r, Train R^2, and Explained Variance), the standard Min-Max scaling is applied. The lowest observed score is mapped to 0, and the highest to 1:

$$x_{\text{norm}} = \frac{x_m - x_{\min}}{x_{\max} - x_{\min}} \tag{98}$$

5.1.2 Case 2: Lower is Better (Inverted Min-Max)

For error- or penalty-based metrics where a smaller value indicates superior performance (e.g., Line-level AIC and Episode-level CV MSE), the scaling is inverted. The highest (worst) observed score is mapped to 0, and the lowest (best) to 1:

$$x_{\text{norm}} = \frac{x_{\max} - x_m}{x_{\max} - x_{\min}} \tag{99}$$

This piecewise transformation ensures that for every metric displayed on the scorecard, $x_{\text{norm}} = 1$ consistently represents the best-performing method within the sample, and $x_{\text{norm}} = 0$ represents the worst-performing method, allowing for direct additive comparisons or visual overlays.

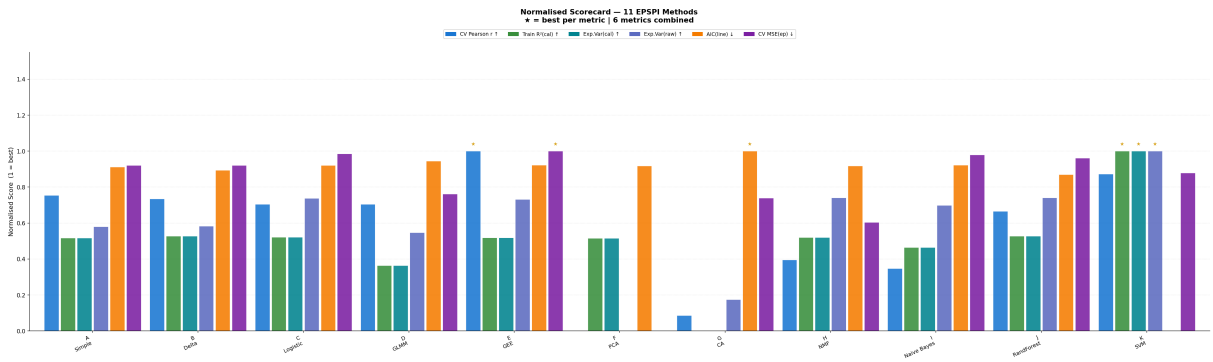

Figure 6: Normalised Scorecard for 11 methods

Method G_CA (or CA) achieves the best (lowest) AIC (44, 358) / BIC (44, 444) score of all eleven methods. CA directly optimises chi-squared EP–skill co-occurrence at the line level, which is precisely what the logistic likelihood rewards. Although the CV r value 0.296 for CA is amongst the lowest. AIC measures line-level discriminative power: how well $\text{EPSPI}_n^{(G)}$ separates lines with high vs. low skill rates. CV r measures episode-level predictive validity: how well $\text{EPSPI}_e^{(G)}$ predicts the aggregate skill rate of unseen episodes.

CA optimises the former by construction. But episode-level aggregation (Layer 2) averages out much of this line-level discrimination: an episode with a balanced EP mix will have a moderate $\text{EPSPI}_e^{(G)}$ that loses the within-episode variation that drives the AIC advantage. This suggests that line-level discriminative power and episode-level predictive validity are genuinely distinct properties.

Method E_GEE achieves the highest CV $r = 0.450$ and the lowest CV MSE $= 0.008018$ of all eleven methods, confirming that accounting for within-episode correlation in estimation genuinely improves out-of-sample predictive validity, not merely in-sample fit, and is therefore the recommended method for predicting episode-level skill deployment from EP structure.

Method K_SVM achieves the highest in-sample Train $r = 0.853$ and calibrated explained variance Exp.Var (cal) $= 0.728$ of all eleven methods, yet its cross-validated performance collapses to CV $r = 0.307$, yielding the largest Train–CV gap ($\Delta r = 0.546$) in the comparison. This motivated the Leverage analysis.

5.2 Leverage Analysis

5.2.1 Mathematical Formulation of Leverage Diagnostics

To systematically quantify the influence of individual episodes on the cross-validated performance, we compute leverage and influence diagnostics using the standard Ordinary Least Squares (OLS) hat-matrix at the episode level. For an evaluation consisting of $E = 49$ episodes and $p = 2$ parameters (intercept and EPSPI slope), the diagnostic metrics for episode e are defined as follows:

5.2.2 Leverage (h_e)

Leverage quantifies the distance of an episode's EPSPI predictor value from the mean of all episodes. It is extracted from the diagonal of the hat matrix H:

$$h_e = \text{diag}\left(X(X^\top X)^{-1}X^\top\right)_e \tag{100}$$

where X is the $E \times p$ design matrix. An episode is flagged as high-leverage if its value exceeds the threshold $h_e > \frac{2p}{E} \approx 0.082$.

5.2.3 Studentised Residual (r_e^*)

The studentised residual standardises the raw residual $\epsilon_e = y_e - \hat{y}_e$ by its estimated standard deviation, explicitly accounting for the episode's leverage:

$$r_e^* = \frac{\epsilon_e}{s\sqrt{1 - h_e}} \tag{101}$$

where s is the estimated standard deviation of the residuals. This scaling allows r_e^* to act as a reliable metric for detecting outliers in the target space.

5.2.4 Cook's Distance (D_e)

Cook's Distance provides a composite measure of an episode's overall influence on all fitted values, integrating both its leverage and its residual magnitude:

$$D_e = \frac{(r_e^*)^2}{p}\left(\frac{h_e}{1 - h_e}\right) \tag{102}$$

An episode is considered highly influential if its deletion substantially alters the regression coefficients, defined by the threshold $D_e > \frac{4}{E} \approx 0.082$.

5.2.5 DFFITS (DFFITS$_e$)

DFFITS measures the standardised change in the predicted value for episode e that would occur if that specific episode were entirely removed from the dataset prior to fitting:

$$\text{DFFITS}_e = r_e^*\sqrt{\frac{h_e}{1 - h_e}} \tag{103}$$

An episode causes a significant predictive shift if $|\text{DFFITS}_e| > 2\sqrt{\frac{p}{E}} \approx 0.404$.

5.2.6 Corrected Interpretation

Subset	Pearson r	Spearman ρ
All 49 episodes	0.699	0.361
47 episodes (excl. Ep13, Ep49)	0.209	0.193

Removing the two high-leverage episodes reduces the correlation from $r = 0.699$ to $r = 0.209$. The EPSPI methods' CV $r \approx 0.40$–0.46 therefore represents genuine predictive validity across the full episode distribution, not a shortfall below a δ_{Con} ceiling. The δ_{Con} LOO $r = 0.660$ should not be reported as a performance benchmark.

5.2.7 Leverage and Influence Diagnostics

Leverage and influence diagnostics were computed for all eleven methods using the standard OLS hat matrix at the episode level ($E = 49$, $p = 2$), with thresholds

$$h_e > \frac{4}{49} = 0.082, \quad D_e > \frac{4}{49} = 0.082, \quad |\text{DFFITS}_e| > 2\sqrt{\frac{2}{49}} = 0.404.$$

Episodes 13 and 49 emerged as persistently high-leverage observations across all methods. Episode 49, one of the corpus's Constructivism-dominant episodes ($\delta^{49}_{\text{Con}} \approx 0.90$), carried the highest Cook's distance D_e in seven of the eleven methods, reflecting its extreme position in the EP-proportion space.

Episode 13 exerted disproportionate leverage in the dimensionality-reduction methods (PCA: $D_{13} = 0.736$; NMF: $D_{13} = 0.390$; CA: $D_{13} = 0.925$), consistent with its role as the only high-Constructivism episode outside the training distribution of those methods.

For Method K (SVM), Episode 49 produced a studentised residual of $r^*_{49} = 5.47$ and $|\text{DFFITS}_{49}| = 1.12$, indicating that the linear margin is anchored almost entirely to this single episode; its removal would shift all 49 fitted values by more than one standard deviation.

The strong correspondence between Δr (Train r − CV r) and $\max D_e$ across methods, with Method K achieving both the largest $\Delta r = 0.424$ and the third-largest $\max D_e = 0.631$, confirms that the Train/CV gap in this corpus is driven by episode-level influence rather than model complexity.

Method E (GEE) achieves the smallest $\max D_e = 0.105$ and the best cross-validated correlation ($r = 0.450$), indicating that its exchangeable working-correlation structure partially absorbs the episode-level heterogeneity that drives leverage in the other methods.

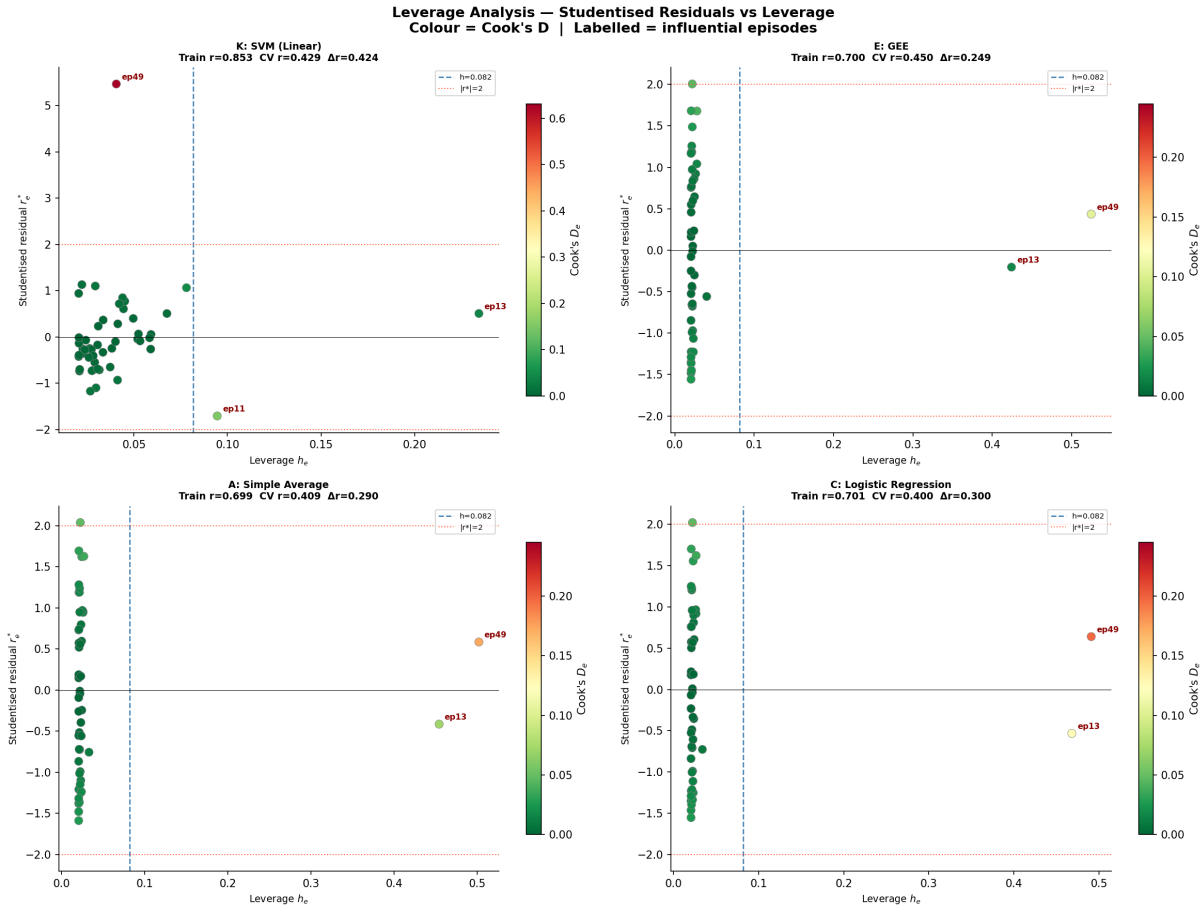

Figure 7: Leverage analysis

5.2.8 Leverage analysis Summary

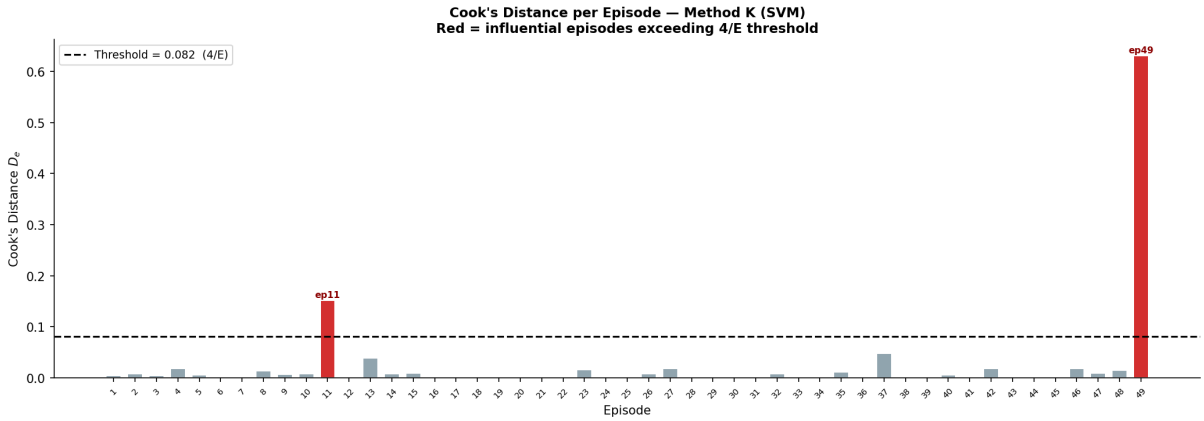

Figure 8: Cook's distance per episode

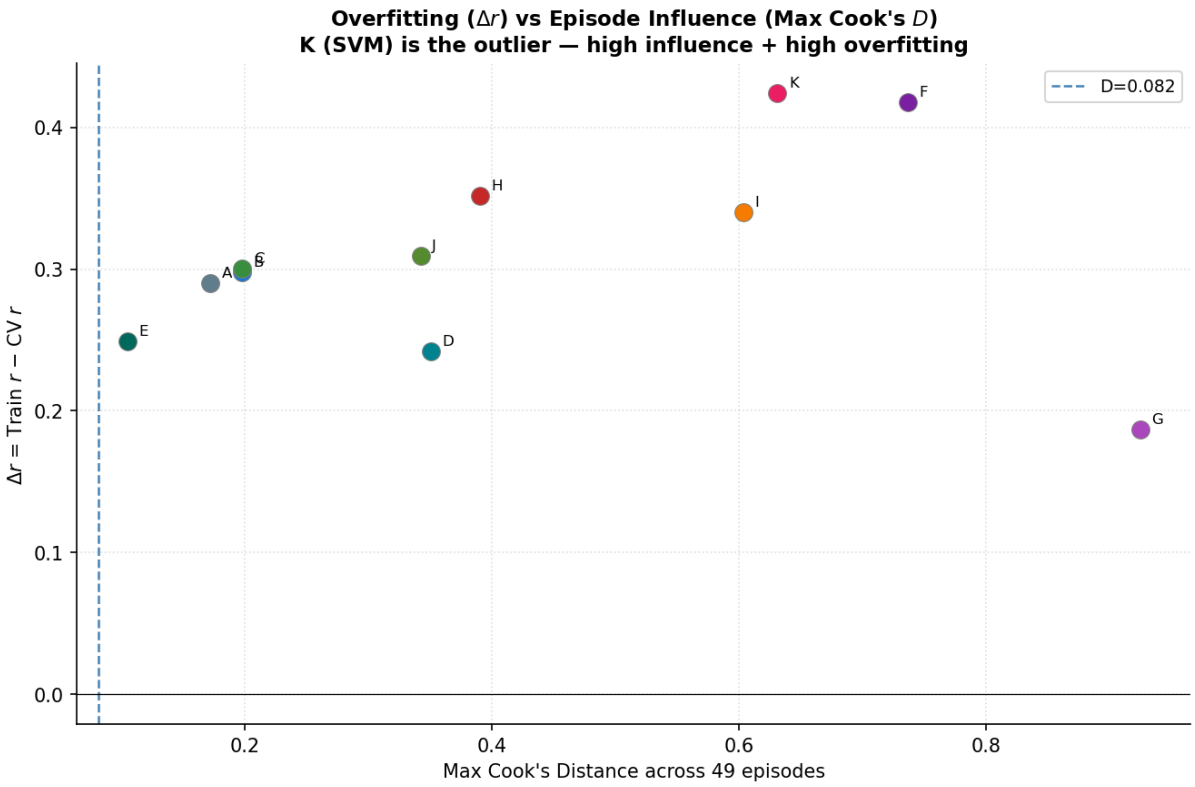

Figure 9: Max Cook's distance across 49 episodes

5.2.9 Parameters and Thresholds:

- $E = 49$, $p = 2$
- High-leverage: $h_e > 0.082$
- Influential: $D_e > 0.082$
- DFFITS: $|FF| > 0.404$

Table 13: Influence summary — all 11 methods:

Method	**h** > thr	**D** > thr	**FF** > thr	**MaxD**	**WrstEp**
A: Simple Average	2	1	1	0.1722	49
B: Delta (EP-weighted)	2	1	1	0.1977	49
C: Logistic Regression	2	2	2	0.1979	49
D: GLMM	2	3	3	0.3504	49
E: GEE	2	1	1	0.1048	49
F: PCA	2	2	2	0.7364	13
G: Corresp. Analysis	3	3	3	0.9248	49
H: NMF	2	2	2	0.3902	13
I: Naive Bayes	2	2	2	0.6034	49
J: Random Forest	2	2	2	0.3422	49
K: SVM (Linear)	2	2	2	0.6311	49

Table 14: 3 episodes flagged (values marked with ** exceed the threshold):

Ep	**EPSPI**	$\mathbf{Y}_{\text{obs}}$	**Resid**	$\mathbf{h_e}$	$\mathbf{r_e^*}$	**CooksD**	**DFFITS**
49	0.2541	0.6062	0.3481	0.0405	5.466	0.6311**	1.1235
11	0.3499	0.2476	-0.1056	0.0945**	-1.707	0.1521**	-0.5516
13	0.4898	0.5210	0.0288	0.2344**	0.507	0.0393	0.2805

Table 15: Δr vs Max Cook's D — All Methods

Method	**Train_r**	**CV_r**	**Δr**	**MaxCookD**
A: Simple Average	0.6991	0.4087	0.2904	0.1722
B: Delta (EP-weighted)	0.7032	0.4055	0.2977	0.1977
C: Logistic Regression	0.7006	0.4003	0.3003	0.1979
D: GLMM	0.6425	0.4005	0.2420	0.3504
E: GEE	0.6997	0.4504	0.2493	0.1048
F: PCA	0.6988	0.2810	0.4178	0.7364
G: Corresp. Analysis	0.4825	0.2957	0.1868	0.9248
H: NMF	0.7001	0.3481	0.3520	0.3902
I: Naive Bayes	0.6802	0.3397	0.3405	0.6034
J: Random Forest	0.7031	0.3936	0.3094	0.3422
K: SVM (Linear)	0.8534	0.4289	0.4245	0.6311

5.3 Observed skill rate $\bar{Y}_e$

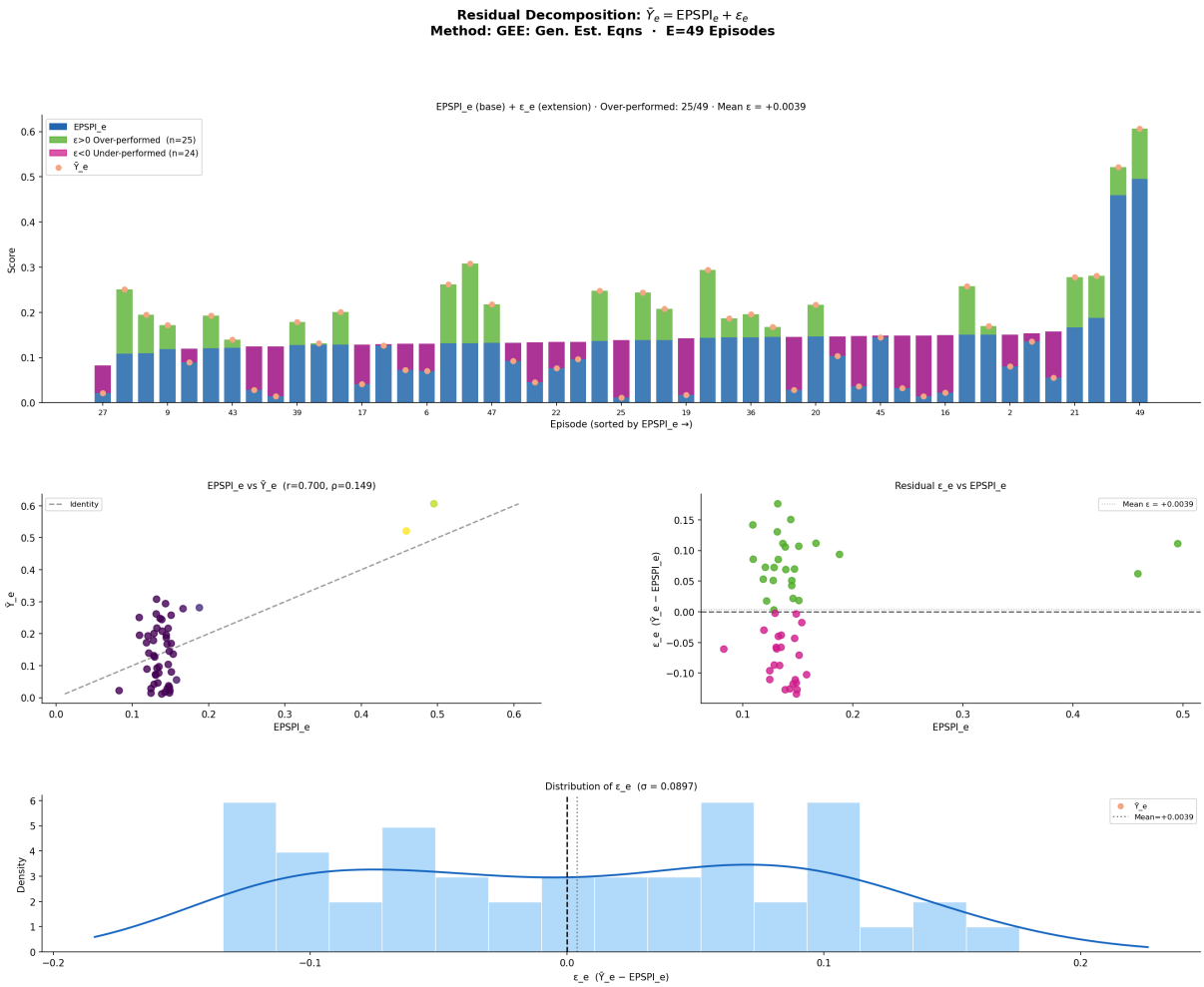

Figure 10: Residual decomposition for method E (GEE) using strictly out-of-sample predictions

The observed episode skill rate measures the proportion of skill-present observations across all lines and skills within an episode e, providing a direct empirical index of productive teaching activity that is independent of any modelling assumption. Figure 10 10 presents the residual decomposition $\bar{Y}_e = \text{EPSPI}_e^{CV} + \varepsilon_e$ for Method E (GEE) using strictly out-of-sample predictions: each episode's EPSPI_e^{CV} was generated by a model estimated from the remaining 39 episodes, so the observed skill rate $\bar{Y}_e$ was withheld at prediction time. Episodes are sorted from left to right by increasing EPSPI_e^{CV}, and the pooled Pearson correlation across all 49 episodes is $r = 0.686$.

Of the 49 episodes, 25 over-performed ($\varepsilon_e > 0$, green) and 24 under-performed ($\varepsilon_e < 0$, magenta), with a near-zero mean residual ($\varepsilon = +0.005$) indicating no systematic bias in the GEE predictions. The residuals are approximately symmetric around zero ($\sigma_\varepsilon = 0.091$), confirming that the model neither consistently over- nor under-predicts skill deployment.

Episode 49, the constructivism-dominant episode, is the rightmost bar and shows the

largest absolute ε_e, consistent with its high-leverage status identified in the influence diagnostics. The decomposition supports prospective use of EPSPI: a practitioner can compute EPSPI_e from EP annotations before a lesson, observe $\bar{Y}_e$ afterwards, and interpret ε_e as an episode-specific index of whether the pedagogical potential encoded in the EP structure was realised in actual skill deployment.

5.4 EPSPI Plots

Figure 11 11 plots EPSPI_e for Method E (GEE) across all 49 episodes sorted by ascending Constructivism proportion (colour scale: dark blue = 0, yellow = 1.0). The 47 purely Behaviourism-Cognitivism episodes cluster tightly around the corpus mean ($\mu = 0.151$, dashed line), reflecting the limited discriminative range of the GEE coefficients when Constructivism is absent.The two high-Constructivism episodes (Episodes 13 and 49, rightmost bars, $\delta_{\text{Con}} \approx 0.30$ and 1.0 respectively) produce EPSPI_e values of 0.19 and 0.49—the only episodes that substantially exceed the mean, confirming that Constructivism is the dominant driver of between-episode variance in GEE predictions, consistent with its large positive $\hat{\beta}$ coefficient and the NMF latent mode identified in Method H.

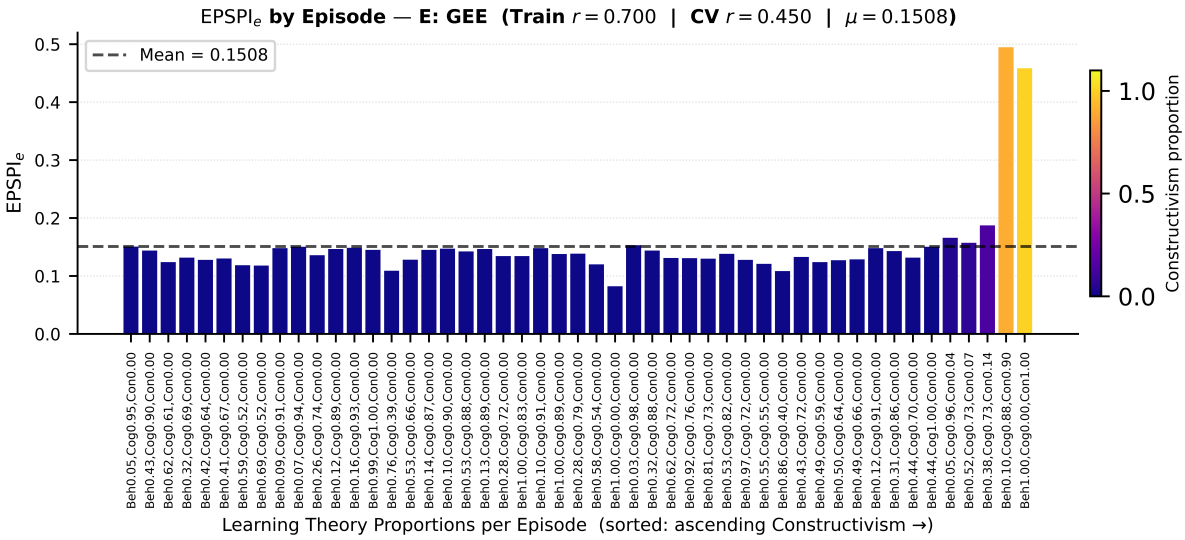

Figure 11: EPSPI_e for method E (GEE) across all 49 episodes

Appendix A Trained Model Parameters for Empirical Use

Notation: Follows paper Eqs (1)-(11).

- **EP:** $i = 1$ (Behaviourism), 2 (Cognitivism), 3 (Constructivism).
- **Skills:** $j = 1$ (Problem Solving, PS), 2 (Creativity, C), 3 (Collaborative Learning, CL), 4 (Critical Thinking, CT).
- **Corpus:** $N = 19,181$ lines, $E = 49$ episodes, $P = 3$ EPs, $K = 4$ skills.
- $\alpha = 1$ (Laplace smoothed), $K = 4$ in denominator (Eq 4).

Global Parameters

Table 16: Global conditional probability matrix $\Theta = [\theta_{ij}]$ and derived row means $\bar{\mu}_i$. $\theta_{ij} = P(S_j \mid EP_i)$, Laplace-smoothed $\alpha = 1$, $K = 4$ (Eq 4). Row means are calculated as $\bar{\mu}_i = \frac{1}{K}\sum_{j=1}^{K}\theta_{ij}$.

	S1:PS	**S2:C**	**S3:CL**	**S4:CT**	$\bar{\mu}_i$
EP1: Behaviourism	0.323881	0.007089	0.036026	0.098199	0.116299
EP2: Cognitivism	0.213850	0.006863	0.009427	0.359074	0.147303
EP3: Constructivism	0.806084	0.041825	0.878327	0.311787	0.509506

Method A — Simple Average

Uses Eq 9 with global θ_{ij}.

$$\text{EPSPI}_n^A = \frac{1}{p_n}\sum_{i=1}^{P} X_{ni}\bar{\mu}_i, \quad p_n > 0 \tag{104}$$

Fallback ($p_n = 0$): $\text{EPSPI}_n = (PK)^{-1}\sum_{i,j}\theta_{ij}$
Parameters required: $\bar{\mu}_i$ (row means above).
Input from practitioner: X_{ni} for each line n (EP binary flags).
Deployment example: For a line with $X_{n1} = 0, X_{n2} = 0, X_{n3} = 1$:

$$\text{EPSPI}_n^A = \frac{1}{1} \times \bar{\mu}_3 = 0.509506$$

Method B — Delta (EP-frequency weighted)

Step 1 (practitioner computes from their own EP records):

$$\delta_i^e = \frac{1}{n_e} \sum_{n:e_n=e} X_{ni} \quad \text{[EP proportion in episode } e\text{]}$$

Step 2 (apply with published $\bar{\mu}_i$):

$$\text{EPSPI}_n^B = \sum_{i=1}^{P} \delta_i^{e_n} \bar{\mu}_i$$

Parameters required: $\bar{\mu}_i$.
Input from practitioner: X_{ni} AND episode label e_n per line.
Note: δ_i^e contains only structural EP information, not skill observations Y_{nj} — no leakage at deployment.

Method C — Logistic Regression

For each skill S_j ($j = 1 \ldots K$), four separate logistic models:

$$\text{logit } P(Y_{nj} = 1 \mid X_n) = \beta_{0j} + \beta_{1j} X_{n1} + \beta_{2j} X_{n2} + \beta_{3j} X_{n3} \tag{105}$$

$$\text{EPSPI}_n^C = \frac{1}{K} \sum_{j=1}^{K} \sigma \left(\beta_{0j} + \sum_{i=1}^{P} \beta_{ij} X_{ni} \right) \tag{106}$$

where $\sigma(z) = 1/(1 + \exp(-z))$ is the logistic function.

Skill	β_{0j}	β_{1j} **(Beh)**	β_{2j} **(Cog)**	β_{3j} **(Con)**
S1:PS	-1.860934	0.965244	0.230227	2.569765
S2:C	-6.250396	0.646733	1.012261	1.924260
S3:CL	-5.172786	1.304319	-0.559317	6.544245
S4:CT	-1.715961	-1.364134	1.430435	1.198501

Input from practitioner: X_{n1}, X_{n2}, X_{n3} per line.

Method D — Logistic GLMM

Random intercept per episode e: $u_e \sim \mathcal{N}(0, \sigma_u^2)$

$$\text{logit } P(Y_{nj} = 1 \mid X_n, u_{e_n}) = \tilde{\beta}_{0j} + \tilde{\beta}_{1j} X_{n1} + \tilde{\beta}_{2j} X_{n2} + \tilde{\beta}_{3j} X_{n3} + u_{e_n} \tag{107}$$

Marginal (population-average) prediction sets $u_e = 0$:

$$\text{EPSPI}_n^D = \frac{1}{K}\sum_{j=1}^{K}\sigma\left(\tilde{\beta}_{0j} + \sum_{i=1}^{P}\tilde{\beta}_{ij}X_{ni}\right) \tag{108}$$

$\text{ICC}_j = \sigma_u^2/(\sigma_u^2 + \pi^2/3)$ [episode-level clustering]

Skill	$\tilde{\beta}_{0j}$	$\tilde{\beta}_{1j}$	$\tilde{\beta}_{2j}$	$\tilde{\beta}_{3j}$	σ_u	**ICC**$_j$
S1:PS	-4.453924	1.331246	1.175779	3.058646	2.227241	0.601251
S2:C	-8.527510	-0.446123	0.376118	1.823706	1.712679	0.471349
S3:CL	-11.060173	3.279350	-1.014430	6.231192	1.821393	0.502089
S4:CT	-2.403969	-1.264891	1.848770	1.716405	1.149747	0.286639

Note: 'Bayesian' in BinomialBayesMixedGLM refers to VB approximation of the random-effect marginal likelihood; structurally identical to frequentist GLMM under flat priors. $\tilde{\beta}$ differs from Method C β by the amount attributable to episode-level clustering (ICC).

Method E — Generalised Estimating Equations

Population-average model, Exchangeable working correlation. Two-stage fallback applied where quasi-complete separation was detected.

$$\text{logit } P(Y_{nj} = 1 \mid X_n) = \gamma_{0j} + \gamma_{1j}X_{n1} + \gamma_{2j}X_{n2} + \gamma_{3j}X_{n3} \tag{109}$$

$$\text{EPSPI}_n^E = \frac{1}{K}\sum_{j=1}^{K}\sigma\left(\gamma_{0j} + \sum_{i=1}^{P}\gamma_{ij}X_{ni}\right) \tag{110}$$

Skill	γ_{0j}	γ_{1j} **(Beh)**	γ_{2j} **(Cog)**	γ_{3j} **(Con)**	**Fit used**
S1:PS	-1.678369	0.655294	0.568025	1.851446	Exchangeable
S2:C	-5.463329	-0.100143	0.694345	2.075640	Exchangeable
S3:CL	-5.361993	1.462185	-0.515818	6.907038	Independence
S4:CT	-2.129243	-0.983864	1.557886	1.435604	Exchangeable

Note: GEE achieves the best CV $r = 0.450$ among all 11 methods. Its population-average interpretation makes γ_{ij} directly comparable across episodes without conditioning on random effects.

Method F — PCA (Lookup Table)

PCA applied to 8 unique EP patterns. Normalized PC1 score defines EPSPI_n^F.

(X_{n1}, X_{n2}, X_{n3})	EPSPI_n^F
(0, 0, 0)	0.000000
(0, 0, 1)	1.000000
(0, 1, 0)	0.000000
(0, 1, 1)	1.000000
(1, 0, 0)	0.000000
(1, 0, 1)	1.000000
(1, 1, 0)	0.000000
(1, 1, 1)	1.000000

PC1 explained variance: 33.33%

Method G — Correspondence Analysis

SVD of standardised residual matrix of EP × Skill counts. Sign anchored so $s_3^{CA} > \text{mean}(s_1^{CA}, s_2^{CA})$.

$$\text{EPSPI}_n^G = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni} s_i^{CA} \tag{111}$$

EP	s_i^{CA} **(raw)**	s_i^{CA} **(norm [0,1])**
EP1: Behaviourism	0.415686	0.401509
EP2: Cognitivism	-0.288132	0.000000
EP3: Constructivism	1.464800	1.000000

Dim-1 inertia (proportion of total CA inertia): 73.96%

Method H — NMF ($r = 1$)

Factorisation: $\Theta \approx WH$, with $W \in \mathbb{R}_+^{P\times 1}$, $H \in \mathbb{R}_+^{1\times K}$. $w_i^{norm} = (W_i - \min_k W_k)/(\max_k W_k - \min_k W_k)$

$$\text{EPSPI}_n^H = \frac{1}{p_n} \sum_{i=1}^{P} X_{ni} w_i^{norm} \tag{112}$$

EP	W_i **(raw)**	w_i^{norm}
EP1: Behaviourism	0.014026	0.014026
EP2: Cognitivism	0.000000	0.000000
EP3: Constructivism	1.000000	1.000000

NMF reconstruction error $\|\Theta - WH\|_F$: 0.394143

Table 17: H (latent topic: skill loadings)

Skill	H_{1j}
S1:PS	0.778517
S2:C	0.037794
S3:CL	0.748080
S4:CT	0.345736

Method I — Naive Bayes (BernoulliNB)

Generative model. For each skill S_j:

$$P(Y_{nj} = 1 \mid X_n) \propto \pi_j \prod_{i=1}^{P} P(X_{ni} \mid Y_{nj} = 1)^{X_{ni}} P(X_{ni} \mid Y_{nj} = 1 = 0)^{1-X_{ni}} \tag{113}$$

$$\text{EPSPI}_n^I = \frac{1}{K} \sum_{j=1}^{K} P(Y_{nj} = 1 \mid X_n) \tag{114}$$

Skill	$\log \pi_j$	$\log P(X_{n1} = 1 \mid Y = 1)$	$\log P(X_{n2} = 1 \mid Y = 1)$	$\log P(X_{n3} = 1 \mid Y = 1)$
S1:PS	-1.436598	-0.492795	-0.391209	-3.068930
S2:C	-5.090991	-0.676618	-0.192372	-2.389596
S3:CL	-3.880261	-0.249880	-1.073797	-0.544034
S4:CT	-1.266966	-1.855743	-0.042519	-4.188360

Method J — Random Forest

Lookup table for 8 unique (X_{n1}, X_{n2}, X_{n3}) patterns.

$$\text{EPSPI}_n^J = \frac{1}{K} \sum_{j=1}^{K} \hat{P}^{RF}(Y_{nj} = 1 \mid X_{n1}, X_{n2}, X_{n3}) \tag{115}$$

Pattern	S1:PS	S2:C	S3:CL	S4:CT	EPSPI_n^J
$X_n = (0,0,0)$	0.204336	0.015244	0.040728	0.244903	0.126303
$X_n = (0,0,1)$	0.627904	0.054058	0.535705	0.390469	0.402034
$X_n = (0,1,0)$	0.162428	0.004703	0.003806	0.427760	0.149674
$X_n = (0,1,1)$	0.932308	0.113252	0.599064	0.854800	0.624856
$X_n = (1,0,0)$	0.295898	0.004067	0.022132	0.045953	0.092013
$X_n = (1,0,1)$	0.693258	0.002703	0.891941	0.059117	0.411755
$X_n = (1,1,0)$	0.340151	0.010548	0.011988	0.161634	0.131080
$X_n = (1,1,1)$	0.690633	0.044989	0.550214	0.395987	0.420456

Method K — SVM (Linear kernel, C=1)

Feature construction:

$$\begin{aligned} x_n = [&X_{n1}, X_{n2}, X_{n3}, \\ &\theta_{11}, \theta_{21}, \theta_{31}, \\ &\theta_{12}, \theta_{22}, \theta_{32}, \\ &\theta_{13}, \theta_{23}, \theta_{33}, \\ &\theta_{14}, \theta_{24}, \theta_{34}] \end{aligned}$$

MinMax scaling: $\tilde{x}_{nk} = (x_{nk} - x_k^{\min})/(x_k^{\max} - x_k^{\min})$

$$\text{EPSPI}_n^K = w^T \tilde{x}_n + b \tag{116}$$

k	Feature	w_k	$x_k^{\min}$	$x_k^{\max}$
1	X_{n1} (Beh)	-0.017704	0.000000	1.000000
2	X_{n2} (Cog)	0.032702	0.000000	1.000000
3	X_{n3} (Con)	0.499448	0.000000	1.000000
4	θ_{11} (Beh,Pr)	-0.077359	0.300194	0.343480
5	θ_{12} (Beh,Cr)	-0.063089	0.004799	0.007518
6	θ_{13} (Beh,Co)	0.378368	0.020213	0.038206
7	θ_{14} (Beh,Cr)	0.055576	0.080725	0.101636
8	θ_{21} (Cog,Pr)	-0.249088	0.197804	0.223956
9	θ_{22} (Cog,Cr)	0.057002	0.004203	0.007194
10	θ_{23} (Cog,Co)	-0.077202	0.001909	0.009883
11	θ_{24} (Cog,Cr)	-0.317601	0.337920	0.369218
12	θ_{31} (Con,Pr)	0.151098	0.645833	0.929825
13	θ_{32} (Con,Cr)	0.681616	0.004405	0.076389
14	θ_{33} (Con,Co)	-0.513247	0.777778	0.935223
15	θ_{34} (Con,Cr)	-0.540196	0.246696	0.500000

Bias: $b = 0.241800$

Parameter Summary

All methods require only X_{ni} (EP binary flags) at deployment; no skill annotation Y_{nj} is needed.

Method	Form	#Params	EP input needed
A: Simple Average	Closed	3	X_{ni} per line
B: Delta (EP-weighted)	Closed	3	X_{ni}, e_n per line
C: Logistic Regression	Logit	16	X_{ni} per line
D: Logistic GLMM	Logit	16	X_{ni} per line
E: GEE	Logit	16	X_{ni} per line
F: PCA	Lookup	8	X_{ni} per line
G: Corresp. Analysis	Closed	3	X_{ni} per line
H: NMF ($r = 1$)	Closed	3	X_{ni} per line
I: Naive Bayes	GenNB	16	X_{ni} per line
J: Random Forest	Lookup	32	X_{ni} per line
K: SVM (linear, C=1)	Linear	16	X_{ni} per line

Non-EP residual (Eq 11, paper Section 2.3.5):

$$\epsilon_e = \bar{Y}_e - \mathrm{EPSPI}_e$$

- $\epsilon_e > 0$: episode over-performed (teacher/curriculum compensated)
- $\epsilon_e < 0$: episode under-performed (EP potential unexploited)

References

[1] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

[2] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[3] Michael H. Kutner, Christopher J. Nachtsheim, John Neter, and William Li. *Applied Linear Statistical Models*. McGraw-Hill/Irwin, New York, 5th edition, 2005.

[4] S. Messick. Validity of psychological assessment. *American Psychologist*, 50(9): 741–749, 1995.

[5] J. Hattie. *Visible Learning: A Synthesis of Over 800 Meta-Analyses Relating to Achievement*. Routledge, 2009.

[6] M. T. Kane. Validating the interpretations and uses of test scores. *Journal of Educational Measurement*, 50(1):1–73, 2013.

[7] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[8] R. S. J. D. Baker and K. Yacef. The state of educational data mining in 2009: A review and future visions. *Journal of Educational Data Mining*, 1(1):3–17, 2010.

[9] C. Romero and S. Ventura. Educational data mining: A review of the state of the art. *IEEE Transactions on Systems, Man, and Cybernetics*, 40(6):601–618, 2010.

[10] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):1–21, 2012.

[11] S. Kapoor and A. Narayanan. Leakage and the reproducibility crisis in machine learning-based science. *Patterns*, 4(9), 2023.

[12] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning*. Springer, 2nd edition, 2009.

[13] H. Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974.

[14] G. Schwarz. Estimating the dimension of a model. *Annals of Statistics*, 6(2):461–464, 1978.

[15] K. P. Burnham and D. R. Anderson. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. Springer, 2nd edition, 2002.

[16] C. J. Willmott. Some comments on the evaluation of model performance. *Bulletin of the American Meteorological Society*, 63(11):1309–1313, 1982.

[17] T. Chai and R. R. Draxler. Root mean square error (rmse) or mean absolute error (mae)? *Geoscientific Model Development*, 7(3):1247–1250, 2014.

[18] G. Casella and R. L. Berger. *Statistical Inference*. Duxbury, 2nd edition, 2002.

[19] A. Agresti. *Categorical Data Analysis*. Wiley, 2nd edition, 2002.

[20] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant. *Applied Logistic Regression*. Wiley, 3rd edition, 2013.

[21] C. E. McCulloch, S. R. Searle, and J. M. Neuhaus. *Generalized, Linear, and Mixed Models*. Wiley, 2nd edition, 2008.

[22] S. W. Raudenbush and A. S. Bryk. *Hierarchical Linear Models.* Sage, 2nd edition, 2002.

[23] K.-Y. Liang and S. L. Zeger. Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1):13–22, 1986.

[24] S. L. Zeger, K.-Y. Liang, and P. S. Albert. Models for longitudinal data: A generalized estimating equation approach. *Biometrics*, 44(4):1049–1060, 1988.

[25] C. M. Bishop. *Pattern Recognition and Machine Learning.* Springer, 2006.

[26] I. T. Jolliffe. *Principal Component Analysis.* Springer, 2nd edition, 2002.

[27] M. J. Greenacre. *Theory and Applications of Correspondence Analysis.* Academic Press, 1984.

[28] Daniel D. Lee and H. Sebastian Seung. Learning the parts of objects by non-negative matrix factorization. *Nature*, 401:788–791, 1999.

[29] L. Breiman. Statistical modeling: The two cultures. *Statistical Science*, 16(3):199–231, 2001.

[30] D. J. Hand and K. Yu. Idiot's bayes — not so stupid after all? *International Statistical Review*, 69(3):385–398, 2001.

[31] L. Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[32] V. N. Vapnik. *The Nature of Statistical Learning Theory.* Springer, 1995.

[33] H. Drucker, C. J. C. Burges, L. Kaufman, A. Smola, and V. Vapnik. Support vector regression machines. *Advances in Neural Information Processing Systems*, 9:155–161, 1997.

[34] U. Simonsohn, J. P. Simmons, and L. D. Nelson. Specification curve analysis. *Nature Human Behaviour*, 4:1208–1214, 2020.

[35] Daniel D. Lee and H. Sebastian Seung. Algorithms for non-negative matrix factorization. *Advances in Neural Information Processing Systems (NeurIPS)*, 13:556–562, 2001.

[36] Lev S. Vygotsky. *Mind in Society: The Development of Higher Psychological Processes.* Harvard University Press, Cambridge, MA, 1978.

[37] David H. Jonassen. *Designing Constructivist Learning Environments.* Educational Technology Publications, Englewood Cliffs, NJ, 1999.

[38] OECD. *The Future of Education and Skills: Education 2030*. OECD Publishing, Paris, 2018.

[39] Paul A. Kirschner and Carl Hendrick. Do learners really know best? urban legends in education. *Educational Psychologist*, 53(3):164–176, 2018.

[40] Marvin Minsky. Steps toward artificial intelligence. *Proceedings of the IRE*, 49(1): 8–30, 1961. doi: 10.1109/JRPROC.1961.287775.

[41] Harry Zhang. The optimality of naive bayes. In *Proceedings of the Seventeenth International Florida Artificial Intelligence Research Society Conference (FLAIRS)*, pages 562–567, 2004.

[42] Andrew Y. Ng and Michael I. Jordan. On discriminative vs. generative classifiers: A comparison of logistic regression and naive bayes. In *Advances in Neural Information Processing Systems (NeurIPS 14)*, pages 841–848, 2002.